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Investigating Sex-Related Bias in Brain MRI using Fair and Explainable Deep Learning

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Investigating Sex-Related Bias in Brain MRI using Fair and Explainable Deep Learning

by

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A THESIS

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Abstract

The increasing use of deep learning in medical imaging has raised concerns about fairness, particularly regarding sex-related biases. This thesis investigates these biases in brain magnetic resonance imaging (MRI) with two main aims: (1) assessing how sex differences influence predictive models and (2) identifying structural brain differences that contribute to these biases to improve fairness and interpretability in artificial intelligence-driven healthcare.

To address the first aim, the thesis examines the impact of sex differences on brain age prediction. By training models on male-only, female-only, and mixed-sex datasets, it reveals significant disparities in prediction accuracy, demonstrating that standard training practices can amplify bias. Saliency maps further show that models prioritize different brain regions depending on dataset composition, reinforcing the need for sex-aware modeling to ensure unbiased and clinically meaningful predictions.

To achieve the second aim, the thesis investigates the structural sex-related differences in the brain by developing a “Sex Differences Map” based on MRI data from multiple large-scale datasets. Post-hoc analysis reveals total intracranial volume as a confounding factor but also shows that the model captures finer cortical and subcortical differences beyond brain size. Key regions influencing sex classification span the inferior and superior frontal lobes, the mesial temporal cortex, and subcortical areas such as the internal capsule, optic chiasm, pons, and lateral cerebellar surfaces.

Together, these studies emphasize the importance of transparency and fairness in artificial intelligence-based medical imaging. Close collaboration with radiologists ensures the clinical relevance of the findings. Future work could apply the sex differences map to identify and mitigate sex biases in disease classification and extend this approach to other demographic variables, contributing to more equitable healthcare solutions.

Preface

This thesis is a culmination of a collaborative effort involving all co-authors, resulting in two publications of which I am the first author. The publications are as follows:

- **M. Dibaji**, N. Gianchandani, A. Nair, M. Singhal, R. Souza, M. Bento, “Studying the Effects of Sex-Related Differences on Brain Age Prediction Using Brain MR Imaging”, *In Workshop on Clinical Image-Based Procedures (pp. 205-214). Cham: Springer Nature Switzerland 2023.* ([Dibaji et al., 2023](#)) ([Code Available](#))
- **M. Dibaji**, J. Ospel, R. Souza, M. Bento, “Sex differences in brain MRI using deep learning toward fairer healthcare outcomes”, *Frontiers in Computational Neuroscience, 18, 1452457. 2024* ([Dibaji et al., 2024](#)) ([Code Available](#))

Portions of chapter 1 and 2 are used with permission from these publications of which I am the second author:

- N. Gianchandani, **M. Dibaji**, M. Bento, M.E. MacDonald, R. Souza, “Reframing the Brain Age Prediction Problem to a More Interpretable and Quantitative Approach”, *Workshop on Interpretable Machine Learning in Healthcare (IMLH) held in conjunction with ICML 2023* ([Gianchandani et al., 2023](#))

My contributions to this paper included executing experimental methodologies for global brain age prediction and the generation as well as interpretation of heatmaps via conventional interpretability techniques. These analyses facilitated comparative assessments with voxel-level brain age prediction, thereby enriching the discussions of the proposed framework.

- N. Gianchandani, **M. Dibaji**, J. Ospel, F. Vega, M. Bento, M.E. MacDonald, R. Souza, “A voxel-level approach to brain age prediction: A method to assess regional brain aging”, *MELBA*, 2024 ([Gianchandani et al., 2024](#))

My role in this work involved conducting experimental investigations into global brain age prediction across cohorts of both healthy and diseased individuals. Additionally, I was responsible for post-processing and standardizing voxel-level predictive maps to facilitate robust cross-comparisons and ensure methodological consistency.

- F. Dehghani, **M. Dibaji**, F. Anzum, L. Dey, et. al., “Trustworthy and Responsible AI for Human-Centric Autonomous Decision-Making Systems” *Journal of Artificial Intelligence Review*, 2024 (under review) ([Dehghani et al., 2024](#))

In this paper, my contributions included authoring sections on Responsible AI main aspects, critically examining challenges associated with its deployment across various domains, and formulating comprehensive guidelines and strategic recommendations for the development of transparent, accountable, and ethically aligned AI systems.

While these papers are not directly included in this thesis, the knowledge and methodologies developed through my contributions to them have significantly informed and complemented the research presented here. Specifically, my role in conducting experiments, analyzing data, and articulating key findings in these studies deepened my understanding of machine learning fairness and explainable AI techniques in medical imaging. Insights gained regarding fairness definitions and explainable AI development, such as generating interpretability heatmaps and voxel-level analyses, were particularly instrumental in shaping the fairness definitions and explainable AI approaches employed in this thesis. These experiences directly supported the conceptual framework and methodological choices made in addressing sex-related biases and enhancing the interpretability of deep learning models presented in subsequent chapters.

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I want to express my deepest gratitude to everyone who has guided, supported, and inspired me throughout this journey. First and foremost, I am incredibly thankful to my supervisor, Dr. Mariana Bento, for believing in me from the start and giving me countless opportunities to grow—both as a researcher and as a person. Her steady support and thoughtful mentorship have shaped this thesis in ways I never imagined.

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On a personal note, my brother, Sina, has been my biggest cheerleader and emotional anchor, always believing in me more than I believed in myself. His kindness and unwavering support carried me through moments of doubt. My heartfelt gratitude goes to my mother, Mina, whose love and support have been the steady force behind all my ambitions. Her encouragement has given me strength, inspiring me to keep pushing forward. I’m also thankful to my sister-in-law, Zahra, whose thoughtful and caring guidance helped me navigate this journey.

Lastly, I am beyond grateful for my best friends, Amir and Mehrnoush, who have stood by me with unconditional support, even from afar. Their encouragement and unwavering belief in me have meant more than words can express.

To all the women fighting for their rights, especially the brave women of my home country, Iran.

For Woman, Life, Freedom.

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Abbreviations

Abbreviation	Definition
AD	Alzheimer’s Disease
ADNI	Alzheimer’s Disease Neuroimaging Initiative (dataset)
AI	Artificial Intelligence
Cam-CAN	Cambridge Centre for Ageing Neuroscience (dataset)
CC359	Calgary-Campinas-359 (dataset)
CNN	Convolutional Neural Network
DL	Deep Learning
FLIRT	FMRIB’s Linear Image Registration Tool
FNIRT	FMRIB’s Non-linear Image Registration Tool
Grad-CAM	Gradient-weighted Class Activation Mapping
KDE	Kernel Density Estimation
LRP	Layer-wise Relevance Propagation
MAE	Mean Absolute Error
ML	Machine Learning
MNI152	Montreal Neurological Institute (MNI) 152 standard brain atlas
MONAI	Medical open network for AI
MRI	Magnetic Resonance Imaging
OASIS	Open Access Series of Imaging Studies (dataset)
ReLU	Rectified Linear Unit

Abbreviation	Definition
SFCN	Simple Fully Convolutional Network
SHAP	SHapley Additive exPlanations
T1-W	T1 Weighted
TIV	Total Intracranial Volume
XAI	Explainable Artificial Intelligence

Chapter 1

Introduction

1.1 Motivation

Artificial Intelligence (AI), which encompasses Machine Learning (ML) and its subset Deep Learning (DL), has demonstrated remarkable potential in healthcare over recent decades, particularly in clinical decision support systems ([Jiang et al., 2017](#)). Among its diverse applications, medical imaging stands out as a pivotal domain, with modalities such as Magnetic Resonance Imaging (MRI) offering critical structural and functional insights for accurate diagnosis and effective treatment planning ([Lundervold and Lundervold, 2019](#)). Traditional ML methods have shown significant promise in this field ([Litjens et al., 2017](#)). However, advancements in DL have further enhanced these capabilities, enabling automatic feature extraction and pattern recognition from complex medical data, thereby improving diagnostic accuracy and generalizability ([Thakur et al., 2024](#)).

Despite these advancements, the deployment of DL in healthcare presents unique challenges due to the high stakes involved. Unreliable or inconsistent DL model performance can have severe consequences, including misdiagnosis and inequitable treatment outcomes ([Vrudhula et al., 2024](#)). Ensuring fairness and mitigating bias are paramount concerns in healthcare DL, alongside the need for transparency. Since DL decisions are meant to assist clinicians in critical decision-making processes, they must be interpretable to enable validation and build trust ([Dehghani et al., 2024](#)).

Given the ethical and clinical importance of fairness in DL, this thesis is motivated by the pressing need to develop models that are not only accurate but also fair, transparent, and trustworthy.

Specifically, it emphasizes understanding and addressing sex-related biases to ensure reliable and equitable diagnostic support for all patients.

1.2 Background

Bias in DL refers to systematic errors produced by algorithms that disproportionately impact specific demographic groups, often delineated by protected attributes such as sex, age, or ethnicity (Suresh and Guttag, 2021). These biases may arise at various stages of the DL pipeline, encompassing data collection, preprocessing, model architecture, and evaluation processes (Suresh and Guttag, 2021). Within critical domains like healthcare, such biases can compromise the reliability, explainability, and fairness of DL systems, ultimately contributing to disparities in diagnosis and therapeutic outcomes (Obermeyer et al., 2019).

Fairness in DL constitutes a complex interplay of ethical, societal, and technical dimensions. It is broadly defined as the absence of prejudice or favoritism toward any individual or group based on inherent or acquired characteristics (Mehrabi et al., 2021). While numerous metrics and algorithmic definitions of fairness have been proposed, each encapsulates distinct aspects of the concept, often leading to inherent trade-offs. Metrics such as demographic parity, equality of opportunity, and calibration fairness are illustrative of the diverse frameworks used to quantify fairness. Consequently, the definition of fairness must be carefully tailored to the contextual and operational constraints of the problem domain, underscoring the need for domain-specific approaches to ethical AI design (Dehghani et al., 2024).

A key area where these challenges manifest is in medical imaging applications, particularly those related to brain health. Brain aging, a natural process characterized by structural and functional changes in the brain, presents a critical context for such applications. These changes include reductions in gray and white matter volumes and alterations in cortical thickness (Raz et al., 2005), influenced by genetic, environmental, and lifestyle factors. Importantly, deviations from typical aging trajectories are often associated with cognitive decline and an increased risk of neurodegenerative diseases. Brain age prediction using MRI data has emerged as a valuable approach to quantifying these deviations, offering biomarkers for early detection and targeted

interventions for age-related cognitive impairments (Gianchandani et al., 2024). An example of neurodegenerative diseases correlated with brain aging is Alzheimer’s disease (AD) which is marked by progressive cognitive decline, memory impairment, and behavioral changes (Fotenos et al., 2005). ML approaches have demonstrated significant potential in identifying patterns of atrophy and functional disruptions associated with AD. However, these models often exhibit sex-specific performance disparities, with women showing faster progression of hippocampus atrophy (Koran et al., 2017) but higher classification accuracy (Klingenberg et al., 2023a). Addressing such disparities is crucial for ensuring equitable diagnostic outcomes.

Sex-related differences in brain MRI data are extensively documented and include notable variations in brain volume, cortical thickness, and subcortical anatomy (Ruigrok et al., 2014). For example, males typically exhibit larger total brain volumes, whereas females demonstrate increased cortical thickness and heightened regional complexity. These anatomical distinctions extend into aging trajectories, with females frequently experiencing accelerated hippocampus atrophy and heightened susceptibility to neurodegenerative changes, such as those observed in Alzheimer’s disease (Lopez-Lee et al., 2024). Such sex-specific differences significantly influence DL model performance, necessitating rigorous methodologies to ensure equitable outcomes across sexes (DeCasien et al., 2022a).

In the realm of medical imaging, sex-related biases in DL applications stem from diverse sources. Imbalanced representation of sexes in training datasets may lead to biased feature extraction processes, thereby degrading model performance for equity-deserving groups (Larrazabal et al., 2020). However, even balanced datasets may fail to alleviate these biases, as intrinsic biological differences and their complex interactions with aging may still drive disparities in model predictions. For instance, studies on Alzheimer’s disease classification have highlighted that even when datasets are balanced, sex-specific variations persist, necessitating deliberate integration of sex as a critical variable in DL model development to ensure robustness and fairness (Klingenberg et al., 2023a).

Explainable AI (XAI) plays a crucial role in improving the transparency of deep learning (DL) models, particularly in high-stakes domains such as healthcare (Saraswat et al., 2022). Techniques like saliency maps, Grad-CAM, and Layer-wise Relevance Propagation (LRP) provide insights into

model predictions by highlighting influential features and regions. These tools help validate the relevance of features and assist in bias detection by revealing differences in feature importance across demographic groups, ultimately improving confidence in AI-driven decisions (Van der Velden et al., 2022). For example, saliency maps have been applied to evaluate the influence of brain regions on sex classification models, uncovering dependencies on specific anatomical features that may propagate biases. Similarly, XAI has been largely used for brain age prediction and disease classification models such as Alzheimer’s, aiding in the identification of regions and patterns that affect predictions (Borys et al., 2023).

1.3 Aims and Contributions

This thesis aims to investigate sex-related biases in DL models applied to brain MRI. Specifically, it examines how sex differences—often relevant for both healthy trajectories and pathology—may cause uneven model performance in diagnostic or prognostic tasks. Given the “black-box” nature of many DL systems, XAI methods are employed to uncover the sources of bias, and collaboration with radiologists provides clinical insights and validation. To accomplish this overarching goal, two main aims are pursued:

1. **Aim 1: Understanding and evaluating sex-related biases in DL models for brain MRI analysis**, with a particular focus on how these biases might influence assessments of non-pathological brain characteristics (e.g., estimating “brain age” in healthy individuals). This involves determining whether standard data handling and training practices inadvertently disadvantage certain subgroups, and clarifying which imaging factors contribute to these biases.
2. **Aim 2: Identify and analyze structural sex differences in the brain**. By leveraging XAI techniques and pinpointing which brain regions are most predictive for sex classification, we gain insight into the morphological cues that drive these biases. This, in turn, provides a foundation for developing more equitable and transparent DL pipelines in neuroimaging.

Under these two overarching aims, this thesis is organized around two major studies, each offering unique insights into sex bias in DL-driven neuroimaging:

1. **Study 1 — “Studying the Effects of Sex-related Differences on Brain Age Prediction using Brain MRI”** Concentrates on Aim 1 by examining how sex-specific versus balanced datasets affect both performance and learned features in brain age prediction. The results reveal how certain data-handling strategies can either magnify or obscure underlying biases and highlight distinct brain regions prioritized by male-only, female-only, or mixed-sex models.

Key Contribution: Demonstrating that experimental design choices-such as training data composition with respect to subgroup representation-led to distinct performance outcomes and interpretability patterns in brain age prediction, highlighting how sex-based differences shape model performance.

2. **Study 2 — “Sex Differences in Brain MRI using Deep Learning toward Fairer Healthcare Outcomes”** Addresses Aim 2 by training a sex classification model on multiple MRI datasets and employing saliency-based XAI to construct a “sex differences map”. This map pinpoints which anatomical regions contribute most strongly to the model’s sex predictions, offering a transparent view of how DL algorithms encode sex-related distinctions.

Key Contribution: Providing an XAI approach to visualize morphological differences between male and female brains, thereby understanding the structural basis of sex-related biases in DL models. The proposed “sex differences map” could be utilized for bias identification for disease scenarios (e.g., Alzheimer’s disease, multiple sclerosis) where sex differences are widely reported.

By focusing on subjects assumed to be healthy, we were able to isolate morphological differences without the confounding influence of pathologies. To achieve robust sample sizes and capture data from diverse acquisition protocols, we aggregated MRI scans from multiple public datasets, allowing the model to learn from a broad and varied cohort. Collaboration with radiologists further strengthened our interpretability analyses, ensuring clinical relevance and reinforcing our confidence in the findings. The studies showed that overlooking sex-based variations may mask critical diagnostic signals or bias risk estimates, whereas systematically identifying and evaluating sex-related biases fosters more robust and transparent DL models. Figure 1.1 demonstrates an overview of aims and studies in this thesis.

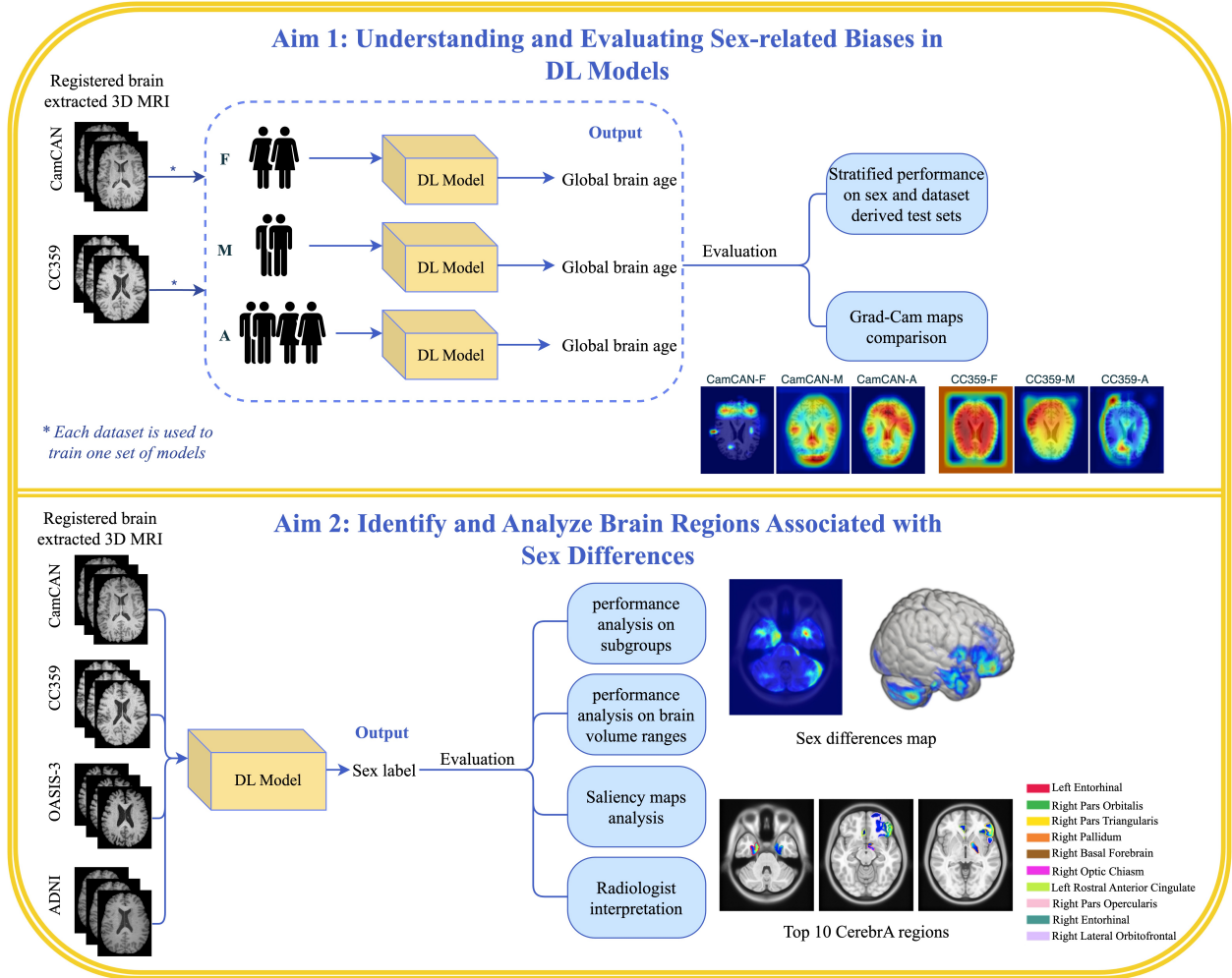


Figure 1.1: Overview of the contributions in this thesis. Aim 1 focuses on identifying and evaluating sex-related biases in deep learning models applied to brain MRI. This includes assessing how sex differences influence model performance and feature attribution in predictive tasks. Aim 2 examines the neuroanatomical basis of these differences by identifying brain regions that contribute to sex classification using explainability techniques and radiologist validation.

1.4 Significance and Impact

This work contributes toward fair DL in medical imaging by investigating a critical area: sex differences in brain MRI. By examining these differences within the training data, experimental design, and final assessments of model outcomes, the study enhances the accuracy and reliability of AI models for both sexes, with an easier translation for clinical application. Furthermore, The methodologies and experiments developed in this research not only address sex as a protected attribute but also offer an approach that can be extended to other demographic attributes, datasets, and medical imaging tasks. This adaptability ensures that the principles and techniques developed

can have a widespread positive impact on the broader field of DL in healthcare.

Moreover, this thesis promotes incorporating fairness considerations into experimental design and model evaluation early in the AI development lifecycle. By leveraging XAI techniques, this research ensures transparency and explainability, fostering trust among clinicians and facilitating the ethical deployment of AI in healthcare. Ultimately, this thesis advances the development of fair, effective, and trustworthy DL systems, promoting equitable healthcare outcomes for diverse patient populations.

1.5 Thesis Outline

This thesis is organized into five chapters:

- **Chapter 1:** Introduction—Provides the motivation, background, research aims, and significance of the work.
- **Chapter 2:** Literature Review and Background—Explores previous research on DL fairness and bias in medical imaging, with a focus on sex differences, and provides relevant background information supporting the research context.
- **Chapter 3:** Study 1—Investigates the effects of sex-related biases and experimental design choices on brain age prediction performance.
- **Chapter 4:** Study 2—Analyzes sex differences in brain MRI using DL XAI techniques and proposes a sex-differences map.
- **Chapter 5:** Conclusions—Discusses the findings of previous chapters, and their implications, and outlines potential future research avenues.

Chapter 2

Literature Review

This chapter surveys core concepts and technical considerations relevant to analyzing sex-related differences in brain MRI with deep learning. It begins by outlining fundamental deep learning techniques and the unique challenges of applying them to MRI data, including preprocessing steps and experimental design choices. Next, it discusses the notion of fairness in deep learning—encompassing how biases can emerge from data or model training—and then examines methods for identifying and mitigating these biases in clinical imaging contexts. The chapter also explores explainability approaches that help illuminate model decision-making processes and detect possible sources of unintended discrimination. Finally, it provides an overview of documented sex-related variations in brain structure and highlights how these differences can influence model performance and healthcare outcomes.

2.1 Deep Learning Overview

Machine Learning (ML) is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed. One common approach is supervised learning, where models are trained on labeled datasets to map inputs to known outputs. This method has been widely used for tasks like image classification and regression. However, traditional machine learning often relies on manual feature engineering, requiring domain expertise to extract meaningful features from raw data. Deep Learning (DL), a subset of ML, overcomes this limitation by using artificial neural networks with multiple layers to automatically

learn complex features directly from raw data. By leveraging large datasets and computational power, DL has achieved remarkable success in tasks such as image classification, object detection, and segmentation, surpassing traditional methods and becoming a key technology in fields like computer vision, natural language processing, and medical imaging (Bengio et al., 2017).

Among the many architectures used in DL, Convolutional Neural Networks (CNNs) are particularly well-suited for image analysis. CNNs consist of several key components that work together to process and understand images. At their core are convolutional layers, which apply filters (kernels) to scan over input images, extracting patterns like edges, textures, and higher-level features (Krizhevsky et al., 2012). Pooling layers then reduce the size of the feature maps, preserving essential information while reducing computational requirements. Activation functions, such as ReLU, introduce non-linearity, enabling CNNs to learn complex relationships in data. Toward the end of the network, fully connected layers aggregate these learned features to make predictions, such as classifying an image or identifying key attributes. Together, these components enable CNNs to learn hierarchical representations, making them particularly effective in medical imaging tasks, where identifying subtle spatial and structural patterns is critical (Haq, 2022).

2.2 Deep Learning Considerations in Brain MRI Analysis

2.2.1 Data Characteristics: Magnetic Resonance Imaging

Magnetic Resonance Imaging (MRI) is a non-invasive imaging technique widely used for generating high-resolution, high-contrast images of soft tissues (Brown et al., 2014). In neuroimaging, T1-weighted (T1w) images are particularly valued for their ability to distinctly highlight gray matter, white matter, and cerebrospinal fluid. These properties make T1w imaging highly effective for structural and volumetric analyses, establishing its role as a cornerstone of neuroimaging research. Its consistent anatomical contrast and reproducibility further contribute to its widespread application in clinical and research contexts.

The effectiveness of DL models in neuroimaging heavily depends on the quality, diversity, and representativeness of the data (Rouzrokh et al., 2022). Datasets should encompass broad demographic variations, including differences in age, sex, and ethnicity, to ensure equitable model

performance and mitigate biases (Puyol-Antón et al., 2022). Additionally, incorporating data from multiple MRI centers introduces valuable variability arising from differences in scanner hardware, acquisition protocols, and imaging conditions, thereby enhancing model robustness. For instance, MRI scanners commonly operate at field strengths ranging from 1.5 to 3 Tesla, with higher field strengths offering superior spatial resolution and improved signal-to-noise ratios (Figure 2.1) (Uğurbil, 2014). Differences between scanner manufacturers (e.g., Siemens, GE, Philips) similarly introduce variations in coil design, reconstruction algorithms, software configurations, and consequently, in image contrast, spatial resolution, and intensity distributions (Figure 2.2) (Souza et al., 2018). (Souza et al., 2018).

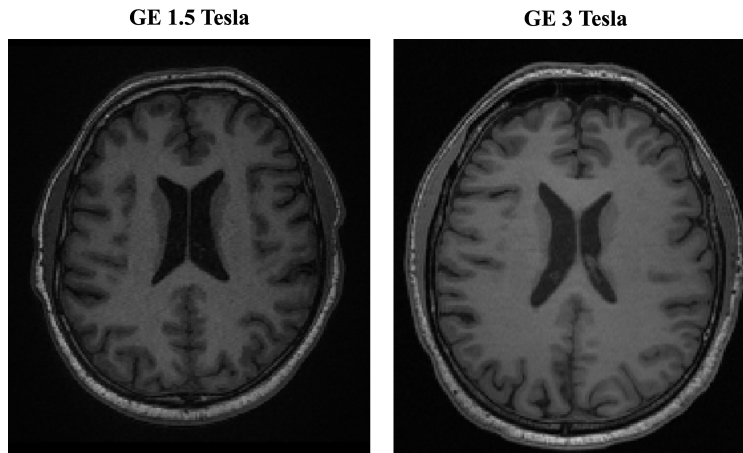


Figure 2.1: Comparison of T1-weighted brain MRI scans from CC359 dataset acquired from healthy male subjects in their 50s at different magnetic field strengths (1.5 Tesla vs. 3 Tesla) using a General Electric (GE) scanner. Higher field strength (3 Tesla) provides improved spatial resolution and contrast, enhancing structural details in brain imaging (Souza et al., 2018).

These inherent variations significantly affect overall image quality and can persist even after standard preprocessing steps, such as intensity normalization. Without careful consideration, deep learning models risk learning scanner- or field-strength-specific artifacts rather than biologically relevant neuroanatomical features. Therefore, training datasets must intentionally represent diverse imaging conditions to ensure models develop invariance to site-related and protocol-specific variations. Such careful dataset construction reduces the risk of overfitting and ensures robust and generalizable model performance, ultimately supporting fairer and more equitable healthcare outcomes.

The use of publicly available datasets is also fundamental to advancing fairness and reproducibility in neuroimaging research. Open access to data facilitates independent validation, enhances compa-

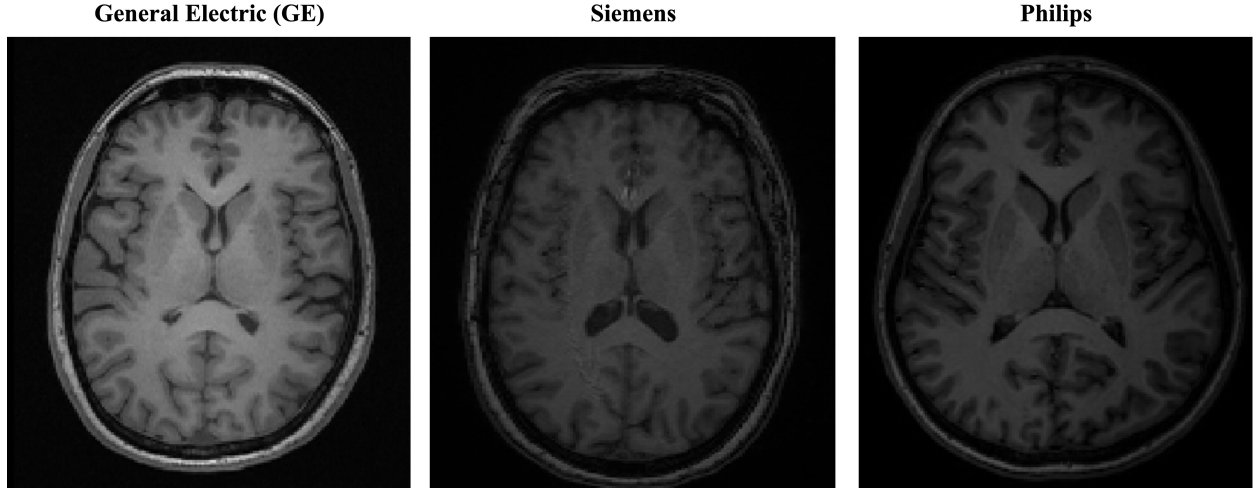


Figure 2.2: Comparison of T1-weighted brain MRI scans from CC359 dataset acquired from healthy female subjects in their 50s using three different MRI scanner manufacturers: General Electric (GE), Siemens, and Philips. Differences in scanner hardware and software configurations contribute to variations in image contrast, intensity, and structural representation, which can impact deep learning models in neuroimaging applications (Souza et al., 2018).

rability across studies, and promotes collaborative progress. By leveraging diverse and accessible datasets, researchers can ensure that their findings are both transparent and generalizable, thereby supporting equitable advancements in the field of DL applied to neuroimaging (Giehl et al., 2024).

2.2.2 Data Processing Techniques

Preprocessing MRI data is a critical step in preparing it for analysis with DL models. Proper preprocessing ensures that the data is standardized and aligned across samples, reducing variability introduced by scanner differences or acquisition protocols.

Skull Stripping Skull stripping or brain extraction entails the removal of non-brain tissues, such as the skull, scalp, and dura, from MRI images to isolate the brain region for analysis. This preprocessing step is integral to ensuring that subsequent analyses focus exclusively on relevant anatomical structures, minimizing the influence of extraneous features. By eliminating irrelevant tissues, skull stripping enhances the accuracy of downstream tasks such as segmentation, classification, and statistical analysis of brain structures, ultimately improving model performance and the reliability of neuroimaging studies (Kalavathi and Prasath, 2016).

Rigid Registration Rigid registration is a method for aligning MRI images by applying transformations constrained to rigid body movements, namely translations and rotations, without introducing any deformations or alterations to the anatomical structures. This process operates within a six-degree-of-freedom framework, comprising three degrees for translation and three for rotation in three-dimensional space. Rigid registration is computationally efficient and robust, making it an essential preprocessing step for aligning images to a common coordinate system. This alignment facilitates meaningful comparisons across subjects or imaging sessions, particularly in scenarios where anatomical variations are minimal.

Non-Linear Registration Non-linear registration, also referred to as deformable or elastic registration, enables the alignment of images by allowing localized deformations to account for significant anatomical differences between subjects or datasets. Unlike rigid registration, this approach involves transformations with a higher degree of freedom, facilitating precise alignment of structures that may vary in shape, size, or position. In this research, non-linear registration was specifically employed to aggregate explainability maps, ensuring their integration across multiple subjects while accounting for individual anatomical variability. Despite these advantages, non-linear registration is computationally intensive and, if not carefully implemented, can introduce noise or artifacts into the processed images that may impact the accuracy of subsequent analyses.

2.2.3 Experimental Design

A well-constructed experimental design is fundamental to ensuring reliable and meaningful results in machine learning studies. Careful planning and execution mitigate biases, maximize generalizability, and provide clear insights into model performance. This section outlines key considerations for experimental design in DL applications, particularly in medical imaging.

Train-Validation-Test Split The train-validation-test split is a cornerstone of experimental design, enabling robust model training, validation, and evaluation. A stratified splitting approach is commonly employed, dividing data while preserving proportions of key labels and factors such as sex, vendor, and magnetic field strength. As discussed in Section 2.2.1, differences in scanner vendor and field strength significantly influence image quality, resolution, and intensity distributions. Such

differences, if not carefully managed, could bias model performance and limit the generalizability of findings. By stratifying splits based on these critical variables, we ensure that the model is exposed consistently to the diverse imaging characteristics present in the dataset, reducing the likelihood of developing dependence on vendor- or field-specific artifacts. This careful stratification thus supports the learning of robust and clinically relevant neuroanatomical features, enhancing the model’s validity and reliability across varied imaging environments. Additionally, while larger training sets typically enhance learning capacity, sufficient validation and test set sizes remain essential for reliable performance assessments. Cross-validation methods can further improve robustness by iteratively validating model performance across multiple splits, although this approach involves higher computational costs and may be less practical for large or computationally demanding datasets (Bengio et al., 2017).

Data Augmentation Data augmentation is a technique used to artificially increase the diversity of the training data by applying transformations to existing samples. This method is particularly advantageous in medical imaging, where annotated datasets are often limited. Augmentation is applied exclusively to training samples, ensuring that the validation and test sets remain representative of real-world data. Common data augmentation techniques in medical imaging include rotation, flipping, intensity scaling, and adding Gaussian noise. These transformations help the model generalize better by exposing it to a broader range of variations, reducing overfitting to the training data. However, the selection of augmentation techniques must align with the dataset characteristics and the task at hand. For instance, intensity scaling may be meaningful for datasets with variable contrast levels, while rotations may be suitable for tasks involving spatially invariant features. Thoughtful augmentation strategies tailored to the problem can significantly enhance model robustness and performance (Bengio et al., 2017).

Hyper-Parameter Tuning Hyperparameters refer to the external configurations of a model that are set prior to training and differ from model weights, which are learned during the training process. Examples of hyperparameters include learning rate, batch size, number of layers, and the choice of activation functions. Hyperparameter tuning aims to identify the optimal configuration for a given task. While grid search and random search are commonly used techniques, exploring all

possible combinations of hyperparameters in DL is often infeasible due to the high computational costs. Instead, practitioners rely on informed initializations and iterative trials to identify suitable hyperparameter values. Bayesian optimization is another advanced approach that can efficiently navigate the hyperparameter space. A well-executed hyperparameter tuning process is essential for maximizing model performance while maintaining computational efficiency (Bengio et al., 2017).

2.2.4 Model Evaluation Metrics and Practices

Evaluation During Training Evaluation during training is typically conducted using a loss function, which quantifies the error between the model’s predictions and the true labels. The loss function guides the optimization process, enabling the model to iteratively improve its predictions. Examples of common loss functions include Mean Absolute Error (MAE) for regression tasks and cross-entropy loss for classification tasks (Bengio et al., 2017).

Evaluation on Test Set Evaluation on the test set assesses the model’s performance on unseen data, providing an unbiased estimate of its generalizability. However, comprehensive evaluation goes beyond reporting a single metric, such as overall accuracy. It is important to evaluate the model’s performance across subgroups, such as demographic or dataset-specific subsets, to ensure fairness and identify any biases. For instance, reporting performance across sex or vendor subgroups may reveal disparities that a general metric might obscure (Bengio et al., 2017).

Task-Specific Metrics For regression tasks, the Mean Absolute Error (MAE) is commonly used to measure the average magnitude of prediction errors. In classification tasks, metrics such as accuracy, balanced accuracy, area under the receiver operating characteristic curve (AUC-ROC), sensitivity, and specificity provide a comprehensive view of model performance. Balanced accuracy is particularly useful in scenarios with imbalanced datasets, as it accounts for class distribution. Sensitivity (true positive rate) and specificity (true negative rate) offer insights into the model’s ability to correctly identify positive and negative cases, respectively. A detailed evaluation using task-relevant metrics and subgroup analyses is critical for assessing both the effectiveness and fairness of the model, ensuring its applicability in real-world scenarios (Bengio et al., 2017).

In conclusion, the development of deep learning models for brain MRI analysis necessitates a

rigorous and multifaceted approach, integrating considerations across data acquisition, preprocessing methodologies, feature extraction strategies, and model evaluation protocols. These components collectively influence the reliability, fairness, and generalizability of deep learning applications in neuroimaging. Figure 2.3 summarizes these critical considerations, providing a structured overview of the fundamental stages of the deep learning pipeline for brain MRI analysis.

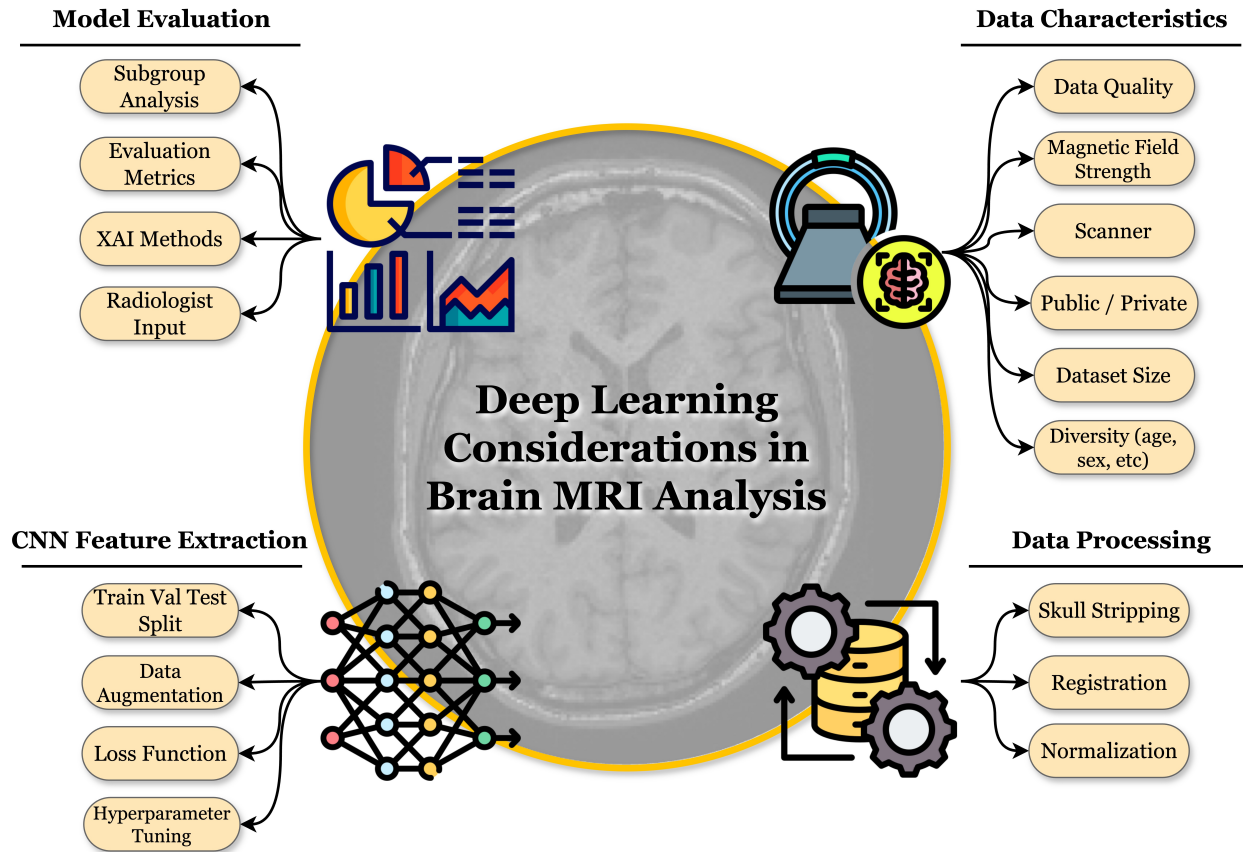


Figure 2.3: Key considerations in deep learning-based brain MRI analysis. These factors are categorized into four main areas: (1) **Data Characteristics**, including aspects such as scanner type, magnetic field strength, dataset size, and demographic diversity; (2) **Data Processing**, covering essential preprocessing steps like skull stripping, registration, and normalization; (3) **CNN Feature Extraction**, highlighting elements such as train-validation-test split, data augmentation, and hyperparameter tuning; and (4) **Model Evaluation**, which includes subgroup analysis, explainable AI (XAI) methods, evaluation metrics, and expert radiologist input. Addressing these aspects is crucial for developing robust and generalizable deep learning models for brain MRI analysis.

2.3 Bias Identification in Deep Learning Models

Bias refers to a systematic error or unintended preference that arises when a model inappropriately relies on protected attributes (e.g., sex, gender, or race) and thereby unfairly disadvantages a particular demographic group (Mehrabani et al., 2021). Bias in DL models arises from multiple sources distributed across the DL lifecycle, encompassing data collection, algorithmic development, and real-world deployment. These biases can profoundly compromise the reliability and fairness of models, particularly in sensitive contexts like healthcare (Suresh and Guttag, 2021).

Biases can be categorized into several key types: representation bias, where training data inadequately represents the target population; aggregation bias, where population-level trends obscure subgroup-specific dynamics; and measurement bias, where inaccuracies in data collection skew outcomes. Historical biases embedded in datasets further perpetuate societal inequities, reinforcing structural discrimination (Mehrabani et al., 2021).

Fletcher et al. (2021) identify additional biases, such as sampling biases during data acquisition and biological biases arising from inherent differences across populations, which may not always be fully addressable. Rajkomar et al. (2018) warn of the risk of amplifying health disparities through biases in patient selection and data representation. Suresh and Guttag (2021) comprehensive framework identifies deployment bias—when models are applied outside their training context—as a significant source of harm.

Bias often arises from larger social inequalities rather than just technical oversights. Passi and Barocas (2019) show that unrealistic assumptions can creep in when broad goals are turned into workable machine learning tasks. This highlights the importance of staying mindful of context and making sure that the system’s end-to-end development remains true to the original aims—right from the first ideas all the way through to real-world deployment.

Mitigating bias in DL models calls for examining each part of the pipeline—from data collection to algorithm deployment—to pinpoint precisely where biases emerge, ensuring that subsequent interventions address these specific vulnerabilities.

Fletcher et al. (2021) recommend several methods to reduce bias, including ensuring representative

data collection processes, leveraging adversarial training techniques, and implementing custom regularization constraints to directly address fairness concerns. [Mehrabi et al. \(2021\)](#) explore advanced techniques such as debiased embedding methods, balancing training dataset distributions, and geometric adjustments that decouple protected and unprotected feature correlations.

Contextual understanding is central to effective mitigation. [Suresh and Guttag \(2021\)](#) emphasize the role of structural knowledge and stakeholder engagement in crafting solutions that align with ethical imperatives. Transparency and inclusivity in the development process not only enhance fairness but also build trust. [Rajkomar et al. \(2018\)](#) advocate for continuous feedback mechanisms involving clinicians and patients to ensure ML models align with healthcare outcomes and undergo periodic impact reassessment.

2.4 Fairness in Deep Learning Models

Fairness in DL encompasses a nuanced interplay of ethical considerations, societal impact, and practical implementation in AI systems. Rather than simply denoting the absence of bias or discrimination, fairness in DL demands the equitable treatment of diverse demographic groups and the production of outcomes that adhere to legal, cultural, and ethical standards. This issue is particularly acute in high-stakes domains such as healthcare, where inequitable outcomes can have life-altering consequences.

[Fletcher et al. \(2021\)](#) describe fairness as requiring an evaluation of algorithmic impacts on demographic groups and selecting fairness definitions that align with specific legal and cultural frameworks. In practice, operationalizing fairness involves translating these fairness concepts into concrete requirements—such as formal metrics, evaluation criteria, and model adjustments—throughout the machine-learning lifecycle. This process can be challenging because fairness metrics often conflict with one another, and promoting equitable outcomes may sometimes require trade-offs against model performance. [Giovanola and Tiribelli \(2022\)](#) expand on this concept, positing that fairness should transcend non-discrimination to include socio-relational dimensions such as fair equality of opportunity and the equal right to justification. This broader perspective underscores the ethical imperative of respecting individuals within their unique socio-economic contexts.

It is important to note that no universal definition of fairness exists. [Pfeiffer et al. \(2023\)](#) describe more than two dozen alternatives, emphasizing how multifaceted fairness is and why context matters. Two well-known definitions include demographic parity, which requires that positive outcomes are distributed equally across groups, and equality of opportunity, which ensures equal true positive rates between groups. Another approach is calibration fairness, which states that people with the same predicted risk (for example, a 10% chance of default or disease) should receive the same decision—such as identical loan terms or medical interventions—regardless of characteristics like race, sex, or age. These varied fairness criteria show the real-world difficulty of juggling competing goals and reconciling conflicts among different fairness definitions.

Within healthcare, [Rajkomar et al. \(2018\)](#) argue for fairness as a vehicle to advance health equity, ensuring that DL systems do not exacerbate disparities among vulnerable groups. Achieving this goal requires a human-centered approach, integrating fairness throughout the DL development and deployment lifecycle.

2.5 Explainability in Deep Learning Models

Explainability in DL models is critical, especially in the context of medical image analysis, where decisions often directly impact patient care. DL models, despite their remarkable predictive performance, are commonly regarded as "black boxes" due to their complex, non-linear architectures that make their decision-making processes opaque. This lack of transparency raises concerns about trustworthiness, reliability, and the ethical deployment of these models in sensitive domains like healthcare. Explainability maps provide a visual explanation of a model's decision-making process, allowing clinicians to validate and trust the predictions. By uncovering how and why a model arrived at a particular decision, these techniques support model refinement, the identification of potential errors, and the discovery of novel biomarkers ([Huff et al., 2021](#)). Additionally, explainability techniques are increasingly vital for meeting regulatory requirements, such as those under the General Data Protection Regulation (GDPR), ensuring the safe and ethical use of AI in clinical settings ([Salahuddin et al., 2022](#)).

Grad-CAM (Gradient-weighted Class Activation Mapping) Grad-CAM is a widely used explainability technique that addresses the limitations of earlier class activation mapping (CAM) methods. It provides localized visual explanations by identifying regions in an image that contribute most to the model’s prediction. Grad-CAM computes the gradients of the target class score with respect to the final convolutional layer’s feature maps. These gradients are globally averaged to generate weights, which are then used to compute a weighted sum of the feature maps. The resulting heatmap highlights the discriminative regions that influenced the model’s decision, with a ReLU activation ensuring only positively associated features are emphasized. Its guided refinement, known as guided Grad-CAM, combines Grad-CAM with guided backpropagation to enhance spatial detail, offering more precise visual cues. Grad-CAM has been employed in tasks such as detecting eye-blink artifacts in MEG images and identifying disease-indicative regions in chest X-rays (Selvaraju et al., 2020).

Saliency Maps Saliency maps are a foundational explainability technique that provides pixel-level importance scores by quantifying the sensitivity of the model’s output to changes in each input pixel. These maps are generated by calculating the gradient of the class score with respect to the input image, and the absolute values of these gradients are visualized as a heatmap. Higher intensities in the heatmap indicate regions that have a stronger influence on the model’s prediction (Simonyan et al., 2013). While saliency maps are simple to compute and widely used, a drawback is their inability to distinguish whether a pixel contributes positively or negatively to a class. This limitation can reduce their interpretive granularity, particularly in binary classification tasks. Despite their susceptibility to noise and lack of robustness in high-dimensional data, their simplicity makes them a valuable tool for understanding the model’s behavior during initial assessments (Huff et al., 2021).

2.6 Sex-related Differences in Brain MRI

Sex encompasses biological characteristics, including chromosomal configurations (e.g., XX or XY), hormonal levels, and reproductive anatomy, typically categorized as male or female. However, as King (2022) emphasizes, sex is not strictly binary. Variations in chromosomal arrangements, gene expression, and hormone levels, along with intersex traits, illustrate that sex exists on a spectrum

rather than within rigid categories. The term “biological sex” has also faced critique for its misuse in invalidating trans identities. Despite this understanding, most datasets do not account for intersex individuals, a limitation that must be acknowledged. Gender, meanwhile, refers to socially constructed roles and identities that vary across cultures and contexts. Differentiating sex from gender is essential in neuroimaging research, as both uniquely influence brain structure and function. Rigorous methodologies are crucial to ensure accurate, equitable interpretations of sex differences (DeCasien et al., 2022b).

Extensive research has documented structural differences in the brain between males and females, even when accounting for total brain volume and other confounding factors. Males generally exhibit larger raw brain volumes, particularly in regions such as the amygdala, hippocampus, putamen, and temporal pole. Conversely, females tend to have greater cortical thickness, increased white matter complexity, and larger regional volumes in the cingulate cortex, superior parietal cortex, and lateral prefrontal cortices (DeCasien et al., 2022b; Ritchie et al., 2018). These differences often exhibit significant overlap between sexes, highlighting that while structural distinctions are measurable, they do not constitute categorical differences. For example, cortical thickness in females has been associated with enhanced connectivity in the default mode network, whereas greater raw brain volume in males has been linked to stronger sensorimotor connectivity.

The biological basis of these structural differences is multifaceted, involving genetic, hormonal, and environmental factors. Hormonal fluctuations, particularly during critical developmental periods such as puberty, play a pivotal role in shaping brain organization and connectivity. These structural variations intersect with sex-biased conditions, including the higher prevalence of autism spectrum disorder and dyslexia in males and the greater incidence of depression and Alzheimer’s disease in females (Ritchie et al., 2018). Notably, faster hippocampus atrophy rates in females with Alzheimer’s disease emphasize the importance of examining sex-related differences in brain aging, as these variations have significant implications for diagnosis, treatment strategies, and understanding disease progression (Klingenberg et al., 2023b).

Although structural differences are well-documented, their translation to functional and behavioral outcomes remains complex and nuanced. Studies frequently demonstrate that substantial structural

differences do not necessarily result in significant disparities in cognitive or behavioral performance. For instance, sex differences in brain volume correlate with specific abilities, such as spatial and verbal skills, but these variations do not account for differences in general intelligence. This suggests the presence of compensatory mechanisms that mitigate functional disparities, highlighting the intricate relationship between brain structure and behavior (Burgaleta et al., 2012b).

Understanding sex-related differences in brain structure also poses critical challenges and opportunities for DL applications in healthcare. DL models trained on brain MRI data must account for these differences to ensure similar performance across sexes. Studies have shown that neglecting sex as a variable can result in biased predictions, potentially leading to disparities in diagnoses and treatment recommendations. Moreover, simplistic methodologies, such as relying on classification metrics like sex prediction accuracy, risk oversimplifying the complexity of brain differences and reinforcing binary categorizations (Sanchis-Segura et al., 2022b). To address these challenges, researchers must adopt rigorous practices, including adjusting for confounding factors such as total intracranial volume, employing diverse and representative datasets, and transparently reporting subgroup performance. These strategies are essential for advancing our understanding of sex differences while fostering fairness and equity in AI-driven medical applications (DeCasien et al., 2022b).

2.7 Sex-related Bias in DL Models

Case Study: Brain Aging Brain age prediction is a machine learning-based technique that estimates an individual’s brain age using neuroimaging data. By comparing the predicted brain age to the actual chronological age, researchers derive the brain age gap or delta, which serves as a biomarker indicating whether a brain appears older or younger than expected for a given age. This delta has been associated with various conditions, including Alzheimer’s disease, mild cognitive impairment, schizophrenia, and multiple sclerosis, and is increasingly recognized as a valuable marker of brain health and cognitive function (Gianchandani et al., 2024). However, recent studies have underscored the need to account for sex-related differences and biases in DL models for brain age prediction to ensure accuracy and fairness.

Sex differences in brain aging are well-documented, with studies highlighting distinct patterns in

regional brain structures and aging trajectories between males and females. For example, [Joo et al. \(2023\)](#) demonstrated that incorporating sex-specific brain characteristics into hybrid DL models improved brain age prediction accuracy. The study found that regional differences, particularly in white matter areas such as the corpus callosum and internal capsule, exhibit distinct aging patterns across sexes. Similarly, [Sanford et al. \(2022\)](#) observed that female brains appeared more youthful than male brains in certain regions, such as the prefrontal cortex, further emphasizing the need to account for sex-specific aging patterns in prediction models.

Alzheimer’s disease (AD) exemplifies how deviations from normal brain aging trajectories can lead to neurodegenerative conditions. AD is characterized by progressive cognitive decline, memory loss, and behavioral changes, primarily caused by the accumulation of amyloid plaques and tau tangles in the brain. Studies have shown that AD-related atrophy patterns align with deviations in brain aging predictions, with significant changes observed in regions such as the hippocampus and the temporal lobe ([Liu et al., 2022](#)). Furthermore, sex differences in AD progression have been reported, with women exhibiting faster rates of hippocampus atrophy and cognitive decline despite comparable levels of pathology. Recent work by [Klingenberg et al. \(2023a\)](#) demonstrated that deep learning classifiers for AD detection exhibit higher performance for women than men, even when trained on sex-balanced datasets, further highlighting the importance of considering sex-specific aging patterns in neurodegenerative disease modeling.

DL models incorporating sex information generally perform better than those that do not. [Usman et al. \(2024\)](#) introduced a multi-task adversarial variational autoencoder (M-AVAE) that explicitly included sex as a variable, enhancing prediction accuracy and reducing bias. This model leveraged adversarial learning to disentangle generic and sex-specific features, addressing biases while maintaining performance. Similarly, [Joo et al. \(2023\)](#) used attention maps to visualize critical brain regions for age prediction, identifying gender-specific contributions and demonstrating that models accounting for sex differences yielded higher accuracy and better explainability.

Explainability methods, such as attention maps and saliency techniques, have been instrumental in identifying the features driving predictions and understanding sex-related biases. [Joo et al. \(2023\)](#) and [Usman et al. \(2024\)](#) employed these methods to pinpoint critical regions influencing brain

age predictions, providing insights into the underlying mechanisms of sex-specific aging patterns. However, the integration of these techniques into broader clinical applications remains limited, with many studies calling for more rigorous approaches to bias mitigation and explainability.

The clinical implications of sex biases in brain age prediction are substantial. Biased models risk exacerbating disparities in diagnostics and treatment recommendations, particularly if they systematically underperform for one sex or fail to capture sex-specific aging patterns. For example, models that neglect sex differences may misinterpret normal aging processes or pathological changes, potentially leading to misdiagnosis or delayed interventions. By incorporating sex as a critical variable and leveraging fairness-aware algorithms, researchers can enhance the reliability and equity of brain age predictions across diverse populations.

In conclusion, while brain age prediction models show great promise for advancing our understanding of neurodegenerative diseases and cognitive aging, addressing sex biases is essential for ensuring their clinical applicability. Incorporating sex-specific features, using robust bias correction methods, and employing explainability techniques are necessary steps toward developing equitable and reliable models that can benefit all individuals.

2.8 Current Challenges

The existing literature highlights the growing importance of addressing fairness and mitigating bias in DL applications within medical imaging. Significant advancements have been made in understanding sex-related differences in brain structure and their implications for diagnostic models. However, critical gaps remain. While prior studies acknowledge the presence of sex-specific biases in areas such as brain age prediction and Alzheimer’s disease classification, these investigations often lack a systematic framework for identifying and quantifying performance disparities across sexes. Moreover, many of the research still tends to emphasize general metrics, such as accuracy, over subgroup analyses, which are crucial for detecting nuanced biases in model predictions.

A notable limitation of the existing body of work is the insufficient use of explainability tools, such as saliency maps or Grad-CAM, to evaluate the features driving DL models’ decisions and to uncover potential biases at the feature level. Similarly, while balanced datasets have been utilized to

address representation bias, this approach alone has proven inadequate for eliminating disparities in model performance. Additionally, much of the existing research is restricted to individual domains or tasks, failing to account for the interplay between different applications, such as brain age prediction and disease classification.

To address these limitations, this thesis presents a comprehensive investigation into sex-related biases in DL models across multiple medical imaging tasks. By employing a rigorous experimental framework that integrates sex-specific and balanced datasets, the studies reveal critical performance disparities and demonstrate the importance of incorporating sex as a key factor in the design and evaluation of DL models. The use of explainable AI methods enables a deeper understanding of model decision-making processes and ensures transparency in identifying bias sources. Furthermore, this research emphasizes the clinical relevance of findings by including diverse datasets and collaborating with domain experts, thereby enhancing the robustness and generalizability of the results.

In bridging the identified gaps, this thesis advances the field by providing an integrated approach that combines bias identification, detailed subgroup analyses, and transparency in model development. The methodologies and findings presented herein contribute to the growing body of knowledge on ethical AI in healthcare and establish a foundation for equitable and transparent DL models applicable to diverse populations.

Chapter 3

Studying the Effects of Sex-related Differences on Brain Age Prediction using brain MR Imaging

This chapter is derived from the following publication:

- **M. Dibaji**, N. Gianchandani, A. Nair, M. Singhal, R. Souza, M. Bento, "Studying the Effects of Sex-Related Differences on Brain Age Prediction Using Brain MR Imaging", *In Workshop on Clinical Image-Based Procedures (pp. 205-214). Cham: Springer Nature Switzerland.* ([Dibaji et al., 2023](#))

This study aligns with the first goal of the thesis—identifying and evaluating sex-related biases in DL models for brain MRI— by systematically examining model performance across different sex-specific datasets. By training and validating models separately on male-only, female-only, and balanced samples, this chapter provides a comprehensive analysis of the impact of sex-based differences on predictive accuracy. Furthermore, Grad-CAM saliency mapping highlights brain regions that are most influential in the model’s decision-making process. This analysis enables a deeper understanding of the distinct features learned by the model under different sex-specific conditions. The findings contribute to the development of more interpretable and equitable machine

learning models, emphasizing the importance of rigorous experimental design and careful subgroup analysis.

3.1 Introduction

ML’s increasing application in predicting brain age through T1-weighted MRI scans has become a key area of focus (Peng et al., 2021), related to brain development, cognitive decline, and neurodegenerative diseases Franke and Gaser (2019). ML models often aim for a ‘global’ brain age index, reflecting brain maturity and serving as a biomarker to assess structural changes and aging Cole and Franke (2017); Wang et al. (2019a). Exploring what these models learn and the significant brain regions they identify may offer insights into individual brain variations.

A crucial aspect to consider when developing these models is the sex differences in brain volumes, which could significantly influence their predictions. Existing literature has demonstrated considerable sex-related differences in the total and regional brain volumes, including Gray Matter, White Matter, and Cerebrospinal Fluid Gur et al. (1999); Wang et al. (2019b). These differences, and their interaction with age, can have significant impact on cognitive impairment, especially in the elderly Jäncke et al. (2014). Therefore, ensuring our ML models account for these variations and perform equally well across different sex subgroups is crucial for fairness and reliability.

In this paper, we aim to investigate how the performance of brain age prediction models varies across sex subgroups: males only, females only, and balanced datasets. Our goal is to develop accurate and interpretable ML models, while also ensuring their fairness. By understanding how predictions vary across these different subgroups, we seek to offer valuable insights for the development of more reliable and fair ML models, and to promote transparency in ML processes. While focusing on brain age prediction using MRI, our experimental design also can be translatable for other tasks, examining sex differences and dataset variations, enhancing transparency and reliability.

Our experimental design followed a similar approach presented in Larrazabal et al. (2020). We trained and validated different models to perform brain age prediction, considering the following experimental design: a) using only female samples, b) using only male samples, and c) using a balanced mix of both sexes. We also performed experiments using different datasets, with varying

data distributions, evaluating inconsistencies for different populations or equity-deserving groups.

The rest of the paper unfolds as follows: Section 2 covers materials and methodologies. Section 3 outlines our experimental results. Section 4 discusses these findings and their implications. The paper concludes with a summary and potential future work.

3.2 Materials and Methods

3.2.1 Brain MR Datasets

In this study, we utilized the Calgary-Campinas-359 (CC359) (Souza et al., 2018) and the Cambridge Centre for Ageing and Neuroscience (CamCAN) (Shafte et al., 2014; Taylor et al., 2017) datasets. CC359 is a multi-vendor (General Electric (GE), Philips, and Siemens), multi-field strength (1.5 T and 3 T magnetic field strengths) volumetric brain MRI dataset, comprising 359 T1-weighted three-dimensional (3D) volumes. It has balanced sex distribution, with 183 (50.97%) female and 176 (49.03%) male healthy subjects, aged 29 to 80 years. Brain masks are also available for this dataset.

The CamCAN dataset comprises MR images that were collected at a single site (MRC-CBSU) using a Siemens TIM Trio scanner at 3 T magnetic field strength. The dataset is composed of T1-weighted reconstructed brain MR volumes and segmentation masks for certain structures. From the total number of 651 samples, 329 (50.54%) correspond to female subjects, and 322 (49.46%) correspond to male subjects aged 18 to 88 years. Figure 3.1 depicts the distribution of age and sex subgroups in CC359 and CamCAN datasets.

3.2.2 Pre-processing

In this study, the preprocessing pipeline consisted of skull stripping, registration, and intensity scaling. We trained a UNet model using the CC359 images and their corresponding brain masks for the skull stripping step. The UNet architecture was chosen for its effectiveness in handling high-resolution brain MR images. This trained model achieved 97% Dice score on the validation set and was then used to perform brain extraction on the CamCAN dataset, thereby effectively stripping non-brain tissues.

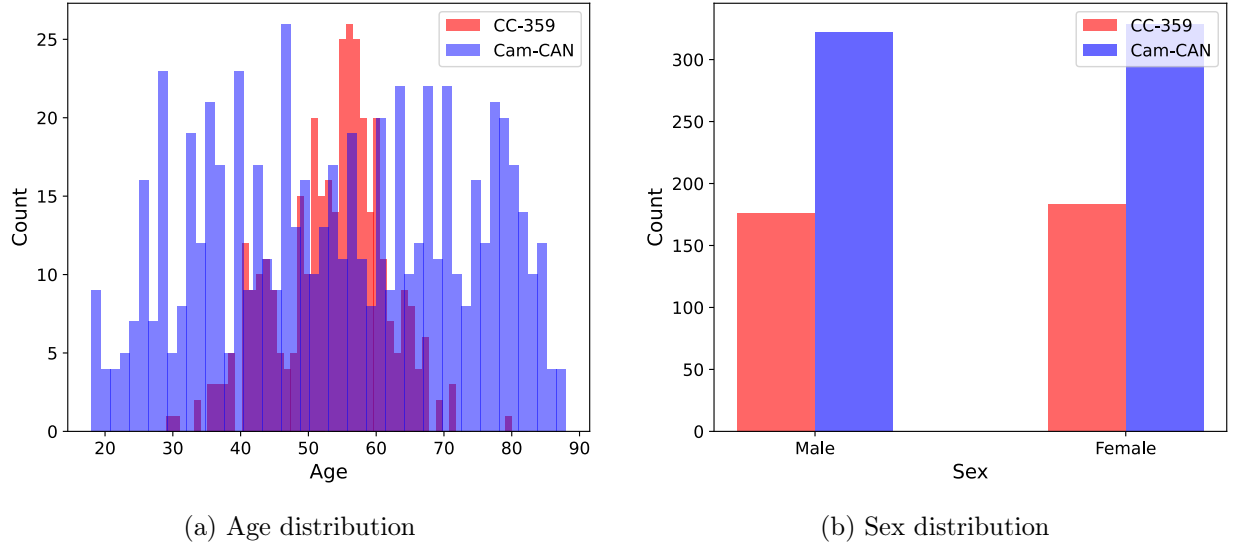


Figure 3.1: Age and sex distribution across the CC359 and CamCAN datasets: (a) The CC359 dataset exhibits an age distribution close to normal and a smaller age range compared to CamCAN. (b) Sex distribution in both datasets reveals a balanced representation of male and female samples.

We registered the skull-stripped brain MR images to the MNI152 standard atlas (Fonov et al., 2009). This registration was performed using FSL’s FLIRT tool (Jenkinson and Smith, 2001; Jenkinson et al., 2002), which allows for a 6 Degrees-of-Freedom rigid registration. This process involves only rotations and translations without distorting the brain’s shape and size. This step ensures that the brain images are comparable and consistent for further analyses. Image intensities were also scaled to fall between 0 and 1.

3.2.3 Brain Age Prediction Task

We utilized a CNN architecture, the Simple Fully Convolutional Network (SFCN) proposed in Peng et al. (2021) for estimating brain age based on 3D T1-weighted images. This model comprises seven convolutional blocks. The initial five blocks down-sample the input after each $3 \times 3 \times 3$ convolutional layer, followed by a $1 \times 1 \times 1$ convolutional block and a classification head. To stabilize the training process, batch normalization is incorporated. The only modification to this model was replacing the classification head with a ReLU-activated linear regressor for brain age prediction.

During the training and validation stages of our study, we employed Mean Absolute Error (MAE) as our loss function. This loss function played a vital role in assessing the disparity between the predicted brain age and the ground truth age labels, evaluating the model’s ability to estimate brain

age accurately.

3.2.4 Grad-CAM explainability

Grad-CAM (Selvaraju et al., 2017) is an explainability method for DL models, especially CNNs. It highlights important regions in an input image by calculating gradients of the target class score with respect to the final convolutional layer’s feature maps. It then creates a heatmap showing regions crucial for the model’s decision, offering insights into its behavior. This helps understand why the model makes certain predictions and is useful for transparency and bias detection in critical applications.

Grad-CAM’s interpretability is limited by coarse localization from lower resolution in deeper layers (McAllister et al., 2020). To overcome this limitation, we averaged maps from both early-stage and final convolutional layers. By blending low-level features like textures with high-level insights, this approach offers a multifaceted interpretation of the model’s reasoning, and enhances the robustness of visualizations by potentially mitigating noise.

3.2.5 Experimental Setting

We established test sets to evaluate the performance of our models through a stratified sampling approach based on vendor and magnetic field, while taking into consideration both sex and dataset. A total of four test sets were constructed in this manner: 30 females from CamCAN (Cam_F); 30 males from CamCAN (Cam_M); 30 females from CC359 (CC_F) and 30 males from CC359 (CC_M).

The remaining samples from each dataset (591 CamCAN and 299 CC359 samples) were used to create three distinct development sets: female subjects only, male subjects only, and larger balanced sets combining both male and female subjects. This resulted in a total of 6 development datasets. Each of these sets was then stratified by vendor and magnetic field strength, and divided into 80% training and 20% validation sets.

This process led to the generation of six separate models. Three were developed using the CC359 dataset, named as CC359-F (trained on female data), CC359-M (trained on male data), and CC359-A (trained on data from all subjects). The remaining three models were trained on the

CamCAN dataset, similarly named as CamCAN-F, CamCAN-M, and CamCAN-A. An illustration of this design is provided in Figure 3.2.

To mitigate overfitting, we employed an augmentation step where 50% of the training and validation samples were randomly subjected to a 15° rotation on-the-fly. Training was carried out using the Adam optimizer with a Mean Absolute Error (MAE) loss function and a batch size of 8. The learning rate was initially set to 0.001, then halved every 10 epochs, across a total of 50 epochs.

For our experiments, we implemented data transforms, and Grad-CAM heat maps using MONAI (Cardoso et al., 2022) which is a robust, open-source platform developed by NVIDIA built upon PyTorch. The University of Calgary Advanced Research Computing (ARC) cluster, specifically gpu-a100 and gpu-v100, were also utilized.

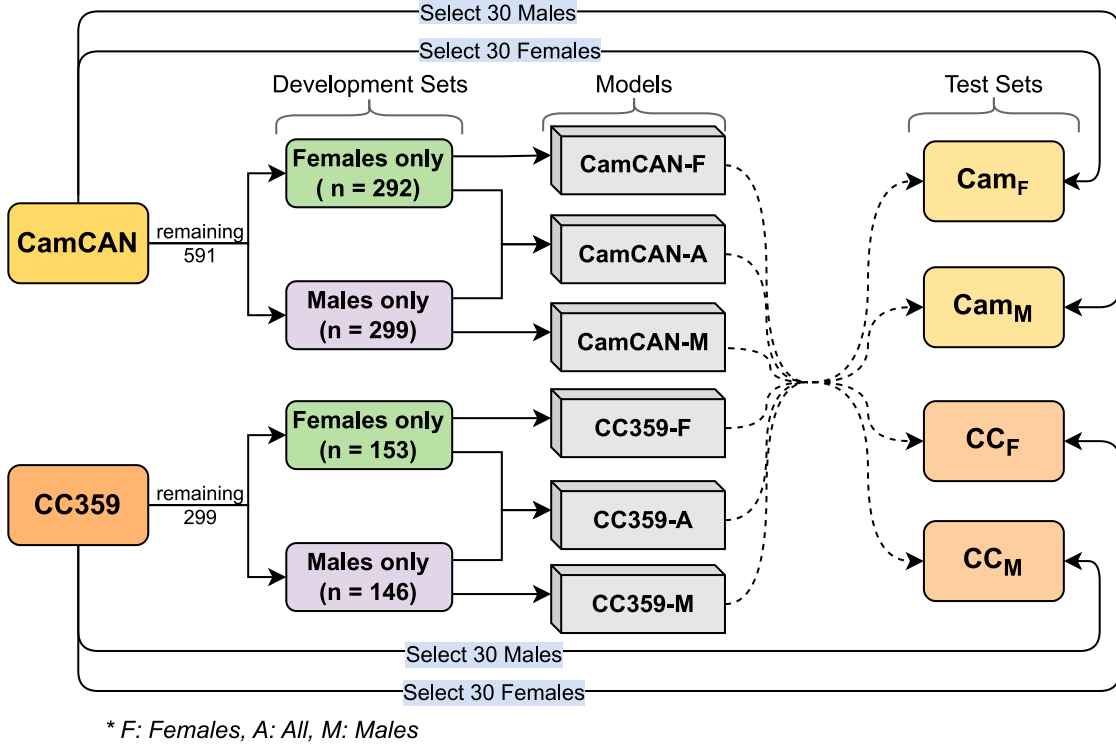


Figure 3.2: Overview of the experimental design. We utilized 3 development sets: Females only, Males only, and All Subjects, extracted from the CamCAN and CC359 datasets separately. These sets were split using a stratified approach based on vendor and magnetic field strength, resulting in training (80%) and validation (20%) subsets. Subsequently, each model was evaluated on four test sets: Cam_F , Cam_M , CC_F , and CC_M

3.3 Results

Table 3.1 provides a summary of the results (MAE) of models trained on CC359 dataset (CC359-F, CC359-M, and CC359-A), evaluated against the defined test sets: Cam_F , Cam_M , CC_F , and CC_M (detailed in section 3.2.5). To compare models performances across sex subgroups regardless of dataset source, we created two aggregated test sets, one each for Males (CC_M and Cam_M combined) and Females (CC_F and Cam_F combined). Table 3.2 presents similar results for models trained using CamCAN dataset (CamCAN-F, CamCAN-M, and CamCAN-A).

In addition, we have produced averaged Grad-CAM maps on each model’s predictions for all test sets, using only the aggregated Females test set for visualization and comparisons purposes (Figure 3.3). These maps represent the mean saliency maps across all samples in this test set. They highlight the regions that each model, on average, considered significant for prediction (Stanley et al., 2022). The same approach was applied to all test sets, yielding similar outcomes across the board. However, we focused on the Females test set to discuss the observed differences.

Table 3.1: Mean Absolute Error (MAE) comparison for three CC359 model: CC359-F (Females Only), CC359-M (Males Only), and CC359-A (All Subjects) across 4 test sets Cam_F , Cam_M , CC_F , CC_M , and 2 aggregated sets combining female (Cam_F and CC_F) and male (Cam_M and CC_M) samples.

Test Sets	CC359-F	CC359-M	CC359-A
Cam_F	17.91 ± 8.864	17.347 ± 9.387	14.355 ± 8.117
Cam_M	16.269 ± 9.452	16.408 ± 8.983	13.804 ± 7.564
CC_F	5.588 ± 5.929	6.47 ± 5.238	5.428 ± 5.291
CC_M	8.311 ± 6.432	7.056 ± 4.883	5.535 ± 4.844
Males	12.29 ± 9.011	11.732 ± 8.61	9.669 ± 7.578
Females	11.749 ± 9.737	11.909 ± 9.346	9.892 ± 8.177

Table 3.2: Mean Absolute Error (MAE) comparison for three CamCAN models: CamCAN-F (Females Only), CamCAN-M (Males Only), and CamCAN-A (All Subjects), across 4 test sets Cam_F , Cam_M , CC_F , CC_M , and 2 aggregated sets combining females (Cam_F and CC_F) and males (Cam_M and CC_M) samples.

Test Sets	CamCAN-F	CamCAN-M	CamCAN-A
Cam_F	7.801 ± 6.186	6.105 ± 4.018	6.366 ± 5.386
Cam_M	7.002 ± 4.69	6.504 ± 4.721	5.763 ± 4.403
CC_F	9.973 ± 6.913	9.571 ± 6.581	9.018 ± 5.963
CC_M	10.049 ± 5.949	12.614 ± 10.144	10.549 ± 7.264
Males	8.526 ± 5.569	9.559 ± 8.481	8.156 ± 6.466
Females	8.77 ± 6.63	7.838 ± 5.721	7.692 ± 5.834

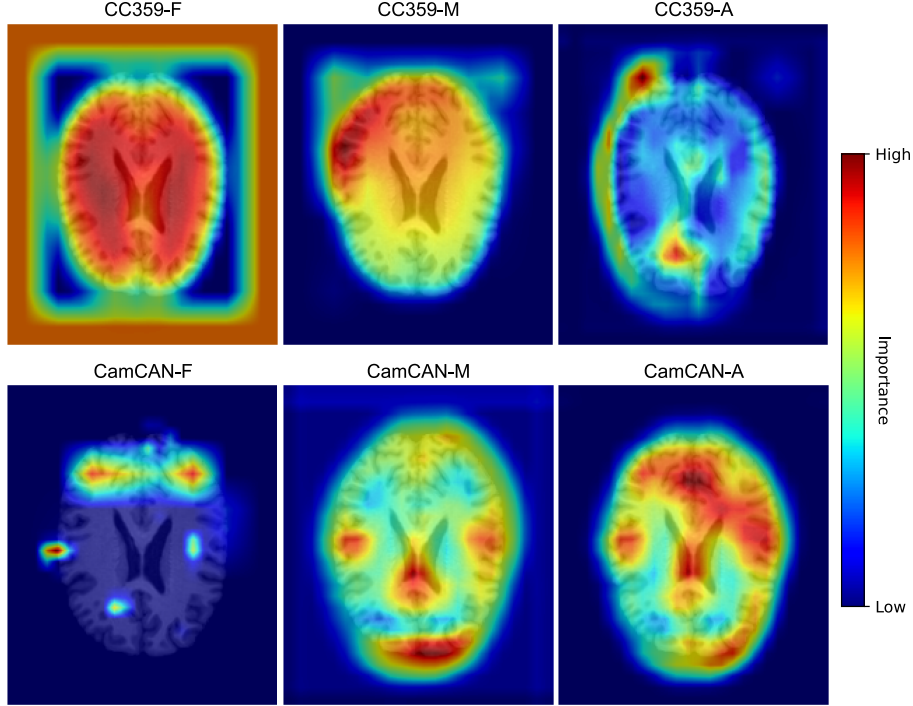


Figure 3.3: Averaged Grad-CAM heatmaps represents a comparative snapshot of significant regions for brain age prediction between models trained on distinct datasets and sex-specific groups, predicted on Females test set (CC_F and Cam_F combined). Although our model is 3D based, we are showing a 2D slice for visualization and comparison purposes (same slice is shown for each model variation.). Models are identified above each map: CC359-F, CC359-M, CC359-A, CamCAN-F, CamCAN-M, CamCAN-A.

3.4 Discussion

The observed results in Table 3.1, which relate to CC359 models, and Table 3.2, which relate to CamCAN models, present a considerable degree of variability. Despite being trained for the same task, using the same architecture, and undergoing the same preprocessing within a closely aligned experimental design, divergent outcomes emerged due solely to differences in the development sets.

Our experiments suggest that models trained on a specific dataset fail to generalize effectively to an external dataset. For instance, in Table 3.1, the MAE of all three models was significantly lower on CC_F and CC_M (sourced from CC359) compared to Cam_F and Cam_M . This pattern is also observed on Table 3.2, although the difference in performance is less significant.

The improved generalizability of CamCAN models can likely be attributed to development set size, as shown in Figure 3.2. While increased diversity in data (e.g., different vendors and magnetic

field strengths) typically improves generalization, this could lead to less accurate predictions if the training samples are insufficient to capture these variations. Therefore, a balance is necessary between dataset size and variability. With increased variability, a larger sample size is required for optimal model function.

The evaluation outcomes on the combined Male and Female test sets demonstrates no consistent trend. Certain models exhibited superior performance on female data, while others were more effective with male data. Interestingly, these performance variations do not seem to be associated with either the origin of the development data or the sample size.

A closer observation of the model performances on sex-specific test sets (especially females) reveals minimal variation among the F, M, and A model variants, as depicted in each table. To better comprehend whether these models were focused on similar features, we used the GradCAM saliency maps as an explainability method. Figure 3.3 shows that even models trained on a specific dataset identified significantly different features when trained using a single-sex subgroup versus a mixed-sex group. This highlights the importance of model’s interpretability to ensure that they are learning the appropriate features.

One model (CC359-F) considered regions outside the brain highly important (red edges in Fig. 3.3), which ideally should not have affected decision-making. For CC359-M, almost all the regions in the brain were relatively important. CC359-A seems focused on smaller regions, however, one of those regions is almost outside of the brain. Similarly, CamCAN-F model is also focused on more specific regions and most of the brain tissue is not important in predicting brain age. Interestingly, CamCAN-M seems to be focused on more specific regions compared to CamCAN-A, despite the fact that CamCAN-A had twice as many training samples and was thus anticipated to learn more specific features.

Our proposal has limitations related to the size of the development sets, potentially affecting the comparisons. However, the performance of models trained on combined males and females was not very different from models trained using only half of the samples (single-sex subgroups). Another limitation is the disregard of age distribution during the split of training, validation, and test sets. Since the CC359 and CamCAN have different age distribution (Figure 3.1.a), an improved

way of splitting training, validation, and test sets should consider the stratified age distribution. Additionally, the use of Grad-CAM, which generates relatively coarse saliency maps may impact the detailed comparison between models and restrict the interpretation of intricate relationships between specific brain regions and predictions. Lastly, the straightforward averaging used to produce an aggregated map for a population presents some challenges, such as potential loss of specificity and alignment complexities. However, these challenges do not diminish the method’s promising potential to improve explainability.

Future work on this topic should delve into two key areas: a more rigorous examination of performance disparity through statistical tests (e.g., Wilcoxon signed-rank test), complemented with the use of more precise saliency maps and a more reliable method for aggregating these maps. This will foster greater confidence in conclusions. Concurrently, efforts must be directed toward designing and optimizing predictive models that specifically address sex-related differences. This dual focus aims to reduce biases and ensure reliable and consistent results across varied populations, strengthening the overall impact of the study.

3.5 Chapter Conclusion

In this study, we examined the influence of sex and dataset variations on brain age prediction. Our findings underscore that experimental design choices significantly shape model performance and feature learning, influencing both reliability and fairness. By demonstrating how sex-specific and mixed-sex datasets yield different predictive outcomes, we highlight the importance of considering demographic variations in deep learning applications. Moreover, the observed disparities in feature attribution emphasize the necessity of explainability methods to ensure trustworthy and explainable outcomes.

Beyond performance variations, our findings raise critical questions about the underlying neuroanatomical differences that may drive sex-based disparities in predictive modeling. While this study primarily focused on brain age estimation, it sets the stage for a deeper exploration of sex differences in brain and their implications for deep learning models. This aligns with the second key aim of this thesis, which is to systematically identify and analyze structural sex differences in the

brain. The next chapter builds upon this foundation by transitioning from a focus on predictive performance to a more detailed investigation of how deep learning models capture and represent sex-related distinctions in brain MRI.

Chapter 4

Sex Differences in Brain MRI using Deep Learning toward Fairer Healthcare Outcomes

This chapter is derived from the following paper:

- **M. Dibaji**, J. Ospel, R. Souza, M. Bento, "Sex differences in brain MRI using deep learning toward fairer healthcare outcomes", *Frontiers in Computational Neuroscience*, 18, 1452457. ([Dibaji et al., 2024](#))

Building on the previous chapter, this study advances the second aim of this thesis by focusing on how neuroanatomical distinctions underlie sex-related biases in DL models for brain. By developing a robust sex classification model and integrating explainability techniques, this work identifies key neuroanatomical features that distinguish male and female brains from a deep learning perspective while systematically assessing potential biases in model predictions. A major contribution of this study is the introduction of a "Sex Differences Map", which provides a structured framework for analyzing how deep learning models make sex-based classifications and highlights the regions most influential in decision-making. The insights gained from this study lay the groundwork for refining bias-aware deep learning techniques, contributing to the development of more equitable and reliable

AI applications in medical imaging.

4.1 Introduction

It is crucial to ensure that DL models perform reliably and consistently across diverse demographic subgroups, such as sex. Fairness in this context refers to the unbiased performance of DL models, where outcomes do not disproportionately favor any specific group, thus avoiding potential discrimination in clinical decision-making (Chouldechova and Roth, 2020). Studying sex differences is particularly important for understanding subtle differences that contribute to unraveling the causes of and treatments for various neurological and neuropsychiatric disorders (Mendrek, 2015). By thoughtfully incorporating sex differences into the development of DL models, we can enhance fairness, ensuring that these models deliver unbiased performance. This approach helps achieve equitable healthcare outcomes for both males and females, reducing the risk of sex-based biases in clinical decision-making.

In our previous study (Dibaji et al., 2023), we assessed the influence of sex on brain age prediction models, observing performance disparities across various sex-specific subgroups and datasets. Similarly, Piçarra and Glocker (2023) also investigated sex and race bias in brain age prediction models, highlighting the need for comprehensive bias assessment in such models. Klingenberg et al. (2023a) found that MRI-based models for Alzheimer’s disease detection performed significantly better for females than for males, even when trained on balanced datasets, underscoring the need for further sex-specific considerations to enhance the fairness and reliability of such algorithms. Although female and male brains are similarly structured, differences exist in the overall brain volume, as well as in cortical and subcortical regions (Williams et al., 2021). Understanding these nuances is beneficial for refining disease diagnosis and comprehending why the prevalence and prognosis of certain neurological diseases differ between males and females (Sanchis-Segura et al., 2022a). Specifically, leveraging this knowledge could optimize DL models, enhancing the performance for both demographic groups.

Several studies have focused on classifying sex from brain medical images using ML methods. A common challenge addressed in these studies is the Total Intracranial Volume (TIV) difference

between males and females, with TIV being, on average, 10-15% smaller in females (Sanchis-Segura et al., 2020). Consequently, studies have proposed methods for careful control of TIV differences between the sexes. One such method is TIV-matching, where males and females of equivalent TIV are matched for the development set (Ebel et al., 2023; Wiersch et al., 2023). With a large sample size, these studies have managed to maintain a high accuracy using traditional ML techniques (Support Vector Machine and Logistic Regression). Another approach involves preprocessing steps, performing spatial non-linear and z-score normalization, which reduces TIV bias to some extent, though not entirely (Ebel et al., 2023).

A key focus of this study is the development of “Sex Differences Map” through saliency analysis, which reveals the critical brain regions involved in distinguishing between male and female brains. These maps provide valuable insights into neuroanatomical distinctions between the sexes and hold potential for guiding the development of more fair and unbiased AI models for healthcare. To create these maps, we developed a DL model for sex classification using four public brain MRI datasets, applying minimal preprocessing, including skull stripping and rigid registration, to preserve brain structure integrity. We utilized a 3D CNN architecture for this task. Instead of controlling for TIV during training, we opted to minimally preprocess the data and later conducted a post-hoc analysis to determine whether the model primarily utilized TIV or successfully identified other critical neuroanatomical features for sex classification. A key component of our methodology was the integration of explainability techniques to identify the brain regions that the DL model deemed important for distinguishing between male and female brains. We assessed these saliency maps both quantitatively, using a region-labeled atlas, and qualitatively, through expert evaluation by a radiologist, ensuring that the model’s focus aligned with relevant neuroanatomical structures. Figure 4.1 provides an overview of the methodology steps.

Different from related works in the literature, we opted not to apply TIV adjustment techniques, such as TIV-matching, during the training phase of this study. While TIV adjustment can help control for volumetric differences between male and female brains, these techniques often require discarding valuable data or performing extensive preprocessing. Such preprocessing can introduce artifacts or lead to the loss of critical neuroanatomical information, particularly in studies focusing on small or intricate brain structures. This loss of detail can potentially obscure relevant features

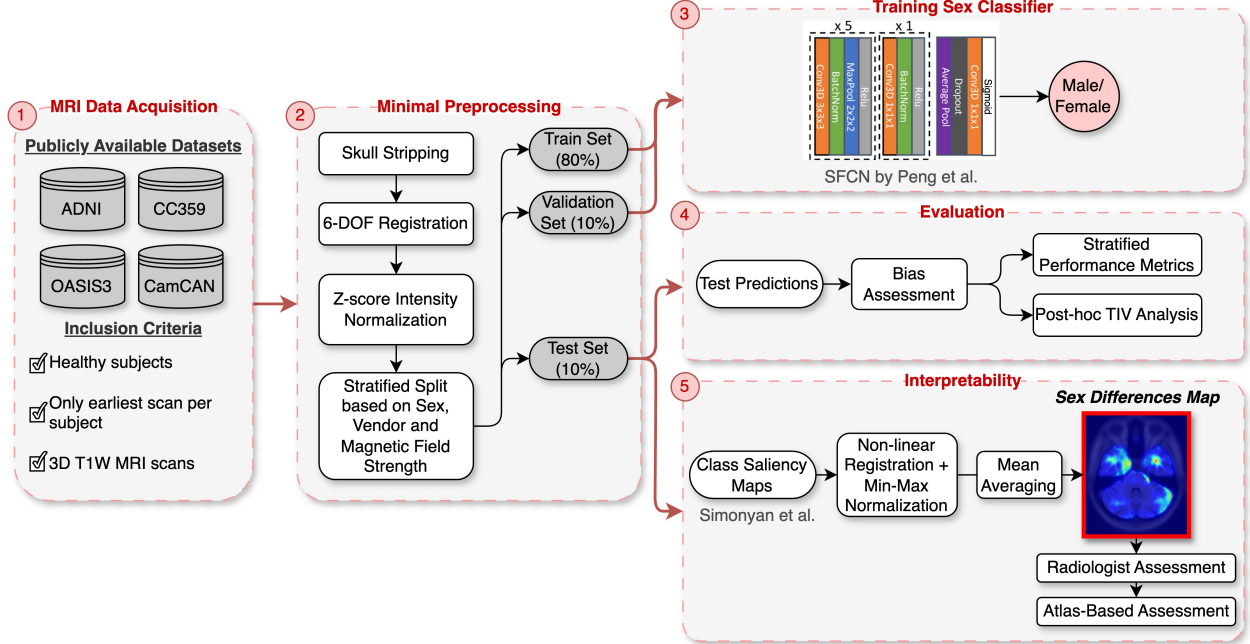


Figure 4.1: Overview of the methodology: (1) Data acquisition from four publicly available datasets (ADNI, OASIS3, CC359, CamCAN). (2) Minimal preprocessing, including skull stripping, rigid registration, and intensity normalization, followed by stratified splitting based on sex, vendor, and magnetic field strength. (3) Training SFCN architecture on the stratified data split. (4) Evaluation with stratified performance metrics and post-hoc TIV analysis to assess bias. (5) explainability using class saliency maps processed into a “Sex Differences Map” and validated through a radiologist and CerebrA atlas assessments.

and limit the overall performance of DL models. Given these concerns, we decided to retain the natural variation in our datasets and perform minimal preprocessing that allowed the model to learn directly from a broader range of brain data. Post-hoc analysis was conducted to assess the model’s reliance on TIV, ensuring that it captured relevant neuroanatomical features beyond simple volumetric differences.

The contributions of this study are as follows: 1) Creation of a “Sex Differences Map” through class saliency analysis, identifying brain regions crucial for distinguishing male and female brains, validated through both quantitative analysis and radiologist evaluation; 2) Adaptation of a 3D CNN architecture for sex classification using publicly available brain MRI datasets with minimal preprocessing to preserve brain structure integrity; 3) Detailed evaluation of model performance, including post-hoc analysis of its reliance on TIV, and detailed performance metrics stratified by datasets, vendors, magnetic field strengths, and age range to assess robustness and identify potential

biases.

4.2 Methodology

4.2.1 Datasets

For the experiments, 3D T1-weighted MR images from four publicly accessible datasets were employed: the Calgary-Campinas-359 (CC-359) dataset (Souza et al., 2018), the Open Access Series of Imaging Studies-3 (OASIS-3) (LaMontagne et al., 2019), the Alzheimer’s Disease Neuroimaging Initiative (ADNI) (Jack Jr et al., 2008), and the Cambridge Centre for Ageing and Neuroscience dataset (Cam-CAN) (Shafto et al., 2014; Taylor et al., 2017). The inclusion of these datasets was driven by the need for healthy control groups, leading us to aggregate data from multiple sources. Additionally, the availability of sex and age information was a critical criterion for dataset selection. Participant demographics in these datasets were determined through self-reporting. In instances where multiple scans were available for a subject, only their earliest scan was used to avoid data leakage. Table 4.1 summarizes the information available for each dataset, including sex, age, vendor, and magnetic field strength. Notably, these datasets represent females and males almost equally.

Table 4.1: Dataset description. Columns from left to right: dataset name, number of subjects (all healthy), sex distribution, age range, MRI vendor, and magnetic field strength.

Dataset	#subjects	Sex (F:M)	Age	vendor	Field (T)
CC-359	359	183:176	29-80	GE, Philips, Siemens	1.5, 3
ADNI	367	186:181	56-90	GE, Philips, Siemens	1.5, 3
OASIS-3	731	416:315	42-95	Siemens	1.5, 3
CamCAN	653	330:323	18-88	Siemens	3

4.2.2 Preprocessing

The pre-processing approach was deliberately kept minimal to standardize the heterogeneous multi-center datasets without significantly altering their original state. Minimal preprocessing preserves the natural neuroanatomical features crucial for tasks like sex classification while ensuring the data is adequately standardized for DL training (Grødem et al., 2024). The specific steps taken, including

brain extraction, rigid registration, and intensity normalization, are standard practices in brain MRI preprocessing (Glasser et al., 2013).

To implement this minimal approach, the initial step in this process involved brain extraction of three datasets using the SynthStrip tool from Freesurfer (skull stripping masks are available for CC359) (Hoopes et al., 2022; Kelley et al., 2024). Subsequently, the brain-extracted images were registered to the MNI152 standard atlas using FSL’s FLIRT tool (Smith et al., 2004). Six-degree-of-freedom rigid registration was chosen to rotate and translate images to a common space without altering individual brain volumes. The processed scans were of size (193, 229, 193), and z-score normalization was applied to image intensities by subtracting the mean and dividing by the standard deviation of the voxel intensities. The processed images have a mean intensity of 0, standard deviation of 1, and a voxel size of 1 mm³. TIV statistics for the raw images in the test set were estimated using the Freesurfer ‘recon-all’ command tool. The estimated TIVs were in mm³ and were divided by 1000 to obtain measurements in milliliters (ml) (Sanchis-Segura et al., 2022a).

4.2.3 Sex Classifier

To perform the sex classification task, we utilized a CNN architecture, the Simple Fully Convolutional Network (SFCN) (Peng et al., 2021). Although this model was proposed for brain age prediction as a classification task, it also showed promising performance in sex classification. This architecture comprises 7 blocks in total: the first 5 blocks each consist of a $3 \times 3 \times 3$ convolutional layer, batch normalization layer, a max pooling layer, and ReLU activation to generate the feature map and reduce the spatial dimension. The sixth block has a $1 \times 1 \times 1$ convolutional layer, a batch normalization layer, and ReLU activation. The last block contains an average pooling layer, a dropout layer (50% during training), and a convolutional layer. The model’s architecture was largely retained as the original model with minor changes: the kernel dimensions of the average pooling layer were set to $6 \times 7 \times 6$ to align with the input size, the number of classes was set to 1 for binary classification, and the softmax output activation was replaced with the sigmoid function to better suit the binary nature of the classification task with one output neuron. Unlike softmax, which is designed for multi-class classification, sigmoid outputs the probability of the instance belonging to the positive class (in this case arbitrarily chosen to be the Male class).

4.2.4 explainability Method

To better comprehend the decision-making process of the DL model, we utilized class saliency maps as proposed by [Simonyan et al. \(2013\)](#). This method involves the computation of the gradient of the class score with respect to the input image layer, using a single back-propagation pass through the image classification network. The gradient indicates the contribution of each voxel to the class score, providing a spatial visualization of the regions within an image that are most important for the model’s decision. We chose this method after testing several alternatives, such as GradCAM and SmoothGrad, because it consistently produced the most detailed and informative saliency maps, crucial for analyzing subtle neuroanatomical differences relevant to sex classification. Additionally, the method is computationally efficient, requiring only a single back-propagation pass through the network. The intensity values in the saliency maps were min-max normalized to range from 0 (least important) to 1 (most important) ([Saporta et al., 2022](#)).

4.2.5 Experimental Design

This study was designed to evaluate the performance of DL, specifically CNNs, for sex classification. We aggregated brain MRI data from four public datasets to explore the sex differences in the brain. The experimental design was carried out in several stages: preprocessing, dataset splitting, training, evaluation, and explainability assessment (Figure 4.1).

Preprocessing steps, including skull stripping, rigid registration, and intensity normalization, were applied to all data prior to the dataset split. These minimal steps were chosen to preserve as much information from the raw data as possible while standardizing the data for model training. The aggregated dataset was divided into three distinct subsets: 80% was allocated for training (1,688 samples), 10% for validation (211 samples), and 10% for testing (211 samples). This division was executed in a stratified manner, ensuring consistent representation of key features—sex, vendor, and magnetic field strength—across subsets. These characteristics were chosen for stratification because they are known to influence MRI data characteristics. Sex differences in brain structure are central to our study, while variations in vendor and magnetic field strength can introduce systematic differences in image quality and signal, which could impact the model’s generalizability and accuracy ([Souza et al., 2018](#)). Within each subset, the sex distribution was 53% female and 47% male, with vendor

distribution of 11% GE, 8% Philips, and 81% Siemens. Magnetic field strengths were represented by 79% at 3 T and 21% at 1.5 T.

To determine the optimal hyperparameters for training the CNN, we conducted preliminary experiments in which we systematically varied key parameters, such as batch size, initial learning rate, and learning rate scheduler. These experiments involved testing a range of commonly used values and assessing their impact on convergence speed, training and validation loss, and overall accuracy. Based on these preliminary results, we selected the following hyperparameters for the final model training: a batch size of 16, an initial learning rate of 0.01, and the Adam optimizer with Binary Cross Entropy as the loss function (Semenov et al., 2019). To further optimize the learning rate during training, we implemented the ‘ReduceLROnPlateau’ strategy, which decreased the learning rate by a factor of 0.1 after five epochs without improvement in validation loss. Training was conducted over 50 epochs.

To mitigate overfitting and improve model generalization, data augmentation was applied during training. Specifically, 50% of the scans were randomly rotated by 15° on-the-fly. This rotation was chosen to introduce variability in scan orientation without distorting key neuroanatomical structures that are crucial for sex classification. Other augmentation techniques, such as random flipping and contrast adjustments, were considered but ultimately not implemented due to their limited impact during preliminary experiments.

The optimized model with the lowest validation loss during the epochs was selected for evaluation on the test set. Performance was assessed using accuracy, balanced accuracy, recall, precision, and F1-score. To further investigate the model’s decision-making process, saliency maps for each correctly classified scan in the test set were generated using the Captum library (Kokhlikyan et al., 2020). Non-linear registration was applied to these maps to align them with the MNI152 standard brain atlas using FSL FNIRT, along with min-max normalization. The processed maps were averaged to identify regions critical for distinguishing between male and female brains. Non linear registration was crucial because the saliency maps were in the native image space of each scan, and variations in brain size and shape across subjects could lead to misalignment during averaging. Therefore, we ensured that corresponding brain regions were aligned across all subjects, allowing for the creation

of a single composite map that accurately reflects sex differences in brain structure from the DL model’s perspective.

The “Sex Differences Map” was first analyzed by a radiologist to ensure that the model’s focus aligned with clinically relevant neuroanatomical structures. After the radiologist’s review, we used the CerebrA atlas to quantify the importance of the specific brain regions identified by the model. The CerebrA atlas provides detailed cortical and subcortical labels for the brain (Manera et al., 2020). For each labeled region, a saliency score was calculated, defined as the proportion of salient voxels within that region (voxels having saliency values greater than 0.1). Additionally, the mean saliency value for each region was computed and then min-max normalized to the range [0,1]. To derive a more representative measure of importance, we calculated a weighted saliency score by multiplying the saliency score by the normalized mean saliency value (Stanley et al., 2022). This approach allowed us to quantify the significance of each brain region in the model’s predictions.

4.2.6 TIV Considerations

TIV is known to be a potential confounding factor in sex classification models. However, in this study, we deliberately chose not to correct for TIV during the training phase. One of the core objectives of our study is to minimize preprocessing steps in order to preserve the integrity of the brain images and enhance the applicability of DL models in clinical settings. TIV correction represents an extensive preprocessing step that might introduce biases or obscure relevant biological information, particularly in small and subtle brain structures. By avoiding TIV correction, we aim to enable the model to learn directly from minimally processed brain images, thereby preserving the natural variations in brain structures that are potentially crucial for accurate sex classification.

Instead of adjusting for TIV during training, we chose to assess its impact on the model’s performance in a post-hoc analysis. This approach allows us to determine the extent to which the model depends on TIV versus other neuroanatomical features, offering a clearer understanding of the model’s reliance on different aspects of the brain data.

For TIV analysis, Freesurfer measurements were used, excluding one unprocessable test sample, resulting in 210 samples. To assess the impact of TIV on model performance, we performed a

post-hoc analysis using TIV quantiles and Kernel Density Estimation (KDE). The test set was divided into three TIV quantiles (each including 70 samples), representing low, medium, and high TIV ranges, to systematically evaluate model performance across different brain sizes (Figure 4.2). This quantile-based approach ensured that each TIV range was equally represented, allowing us to identify any biases the model might have toward extreme TIV values.

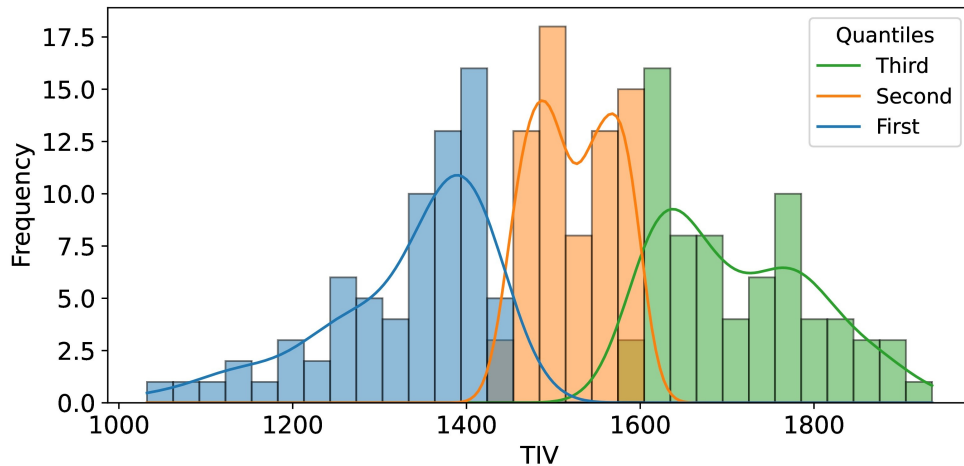


Figure 4.2: The illustration shows the distribution of Total Intracranial Volume (TIV) within three distinct quantiles, each comprising 70 samples. These quantiles have been determined to facilitate an in-depth analysis of model performance variations across varying ranges of TIV.

We also employed KDE to focus the analysis on the TIV range where male and female distributions overlapped the most, i.e., where TIV differences were minimum. A 20% density threshold was applied to exclude the lower-density tails of the distribution, thus emphasizing the regions where the TIV values of both sexes were most similar and TIV was not an overt differentiating factor between sexes. This approach, especially effective in handling non-normally distributed data, focuses on the densest regions, avoiding outliers or less representative data. The threshold was selected after testing several alternatives, finding that 20% provided the most informative results without excluding too much data or including too many extreme cases. This approach helped us isolate the TIV ranges that contributed most to balanced classification performance, minimizing potential sex-based bias.

4.3 Results

The model with the lowest validation loss was evaluated using binary classification metrics. This model achieved an accuracy of 89.5% on the validation set. The overall accuracy on the test set was 87.20%, with subgroup accuracies of 88% for females and 86% for males. Overall AUC-ROC, F1-score, precision, and recall were 93.36%, 86.67%, 86.14%, and 87%, respectively. We conducted a detailed evaluation of model performance across various subgroups based on different datasets, scanner vendors, magnetic field strengths, and age ranges. The performance metrics for each subgroup within these variables are presented in Table 4.2.

Table 4.2: Comparison of performance metrics across different dataset sources, MRI scanner vendors, and magnetic field strengths in the test set.

Variable	Subgroup	Balanced Accuracy	Precision	Recall	F1 Score
Dataset	ADNI	0.863	0.846	0.917	0.880
	CC359	0.851	0.929	0.765	0.839
	OASIS3	0.858	0.731	0.905	0.809
	CamCAN	0.920	0.971	0.868	0.917
Vendor	GE	0.917	1.000	0.833	0.909
	Philips	0.819	0.857	0.750	0.800
	Siemens	0.877	0.855	0.888	0.871
Field	1.5T	0.837	0.818	0.857	0.837
	3T	0.885	0.884	0.873	0.878
Age	< 55	0.889	1.000	0.778	0.875
	55 - 70	0.844	0.788	0.867	0.825
	≥ 70	0.886	0.870	0.930	0.899

Figure 4.3 illustrates the averaged saliency maps for males and females, respectively. This map highlights the brain regions deemed significant by the model for making predictions. It was generated by averaging the saliency maps of accurately classified test samples. Correctly classified samples were utilized in order to identify the most consistent and reliable regions identified by the model. The Spearman correlation between these two maps (females vs. males) was 0.99 ($p < 0.05$), suggesting a very strong positive relationship, indicating that the maps consistently rank the importance of regions in a similar manner.

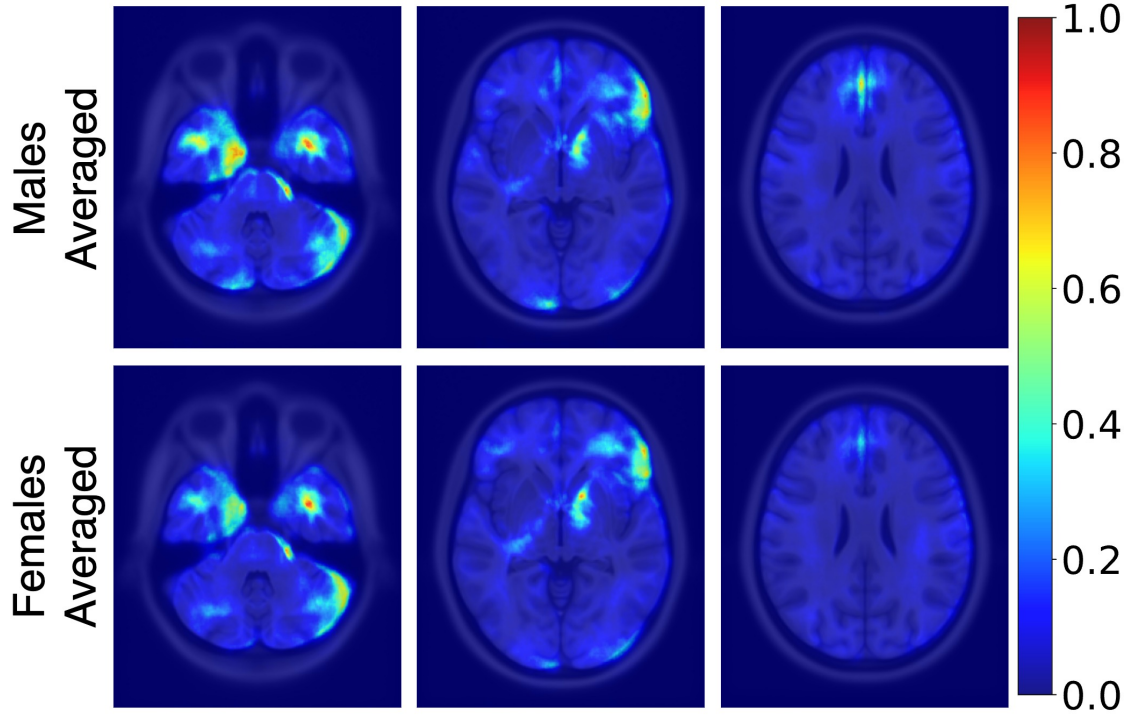


Figure 4.3: Axial MRI slices comparing average saliency maps for correctly classified male (top row) and female (bottom row) subjects. From left to right, Panels (A), (B), and (C) correspond to different axial levels for both male and female subjects. Normalized saliency scores are color-coded from blue (least important, score 0) to red (most important, score 1).

4.3.1 Post-hoc TIV Analysis

In our post-hoc analysis, we observed that the accuracy values across TIV-based quantiles were relatively consistent, ranging from 85% to 89%. However, classification performance varied with TIV; females were classified more accurately at lower TIVs (60 true positives, 1 false positive), while males were more accurately classified at higher TIVs (59 true positives, 1 false positive). Interestingly, the middle TIV quantile showed a similar error rate for both sexes. Table 4.3 summarizes the performance across three TIV quantiles separately for males and females. Balanced accuracy on each TIV quantile highlights the true performance by accounting for the imbalance within the quantiles.

Figure 4.4 shows how predictions varied based on TIV values. It can be observed that most samples in the lower TIV range are classified as female, and most scans in the higher TIV range are classified as male. In other words, as TIV increases, the number of scans classified as male increases. In the middle range of TIV, classification as male versus female is balanced, indicating reduced model bias compared to the higher and lower TIV extremes.

Table 4.3: Model accuracy for sex classification across different Total Intracranial Volume (TIV) quantiles. Each quantile represents a range of TIV values in milliliters, with separate accuracy percentages reported for female and male subjects, as well as the overall balanced accuracy for each quantile.

Quantile	TIV Range	Female Accuracy	Male Accuracy	Balanced Accuracy
First	1032 - 1443	98.4%	33.3%	65.8%
Second	1447 - 1597	87.2%	80.6%	83.9%
Third	1599 - 1935	30.0%	98.3%	64.2%

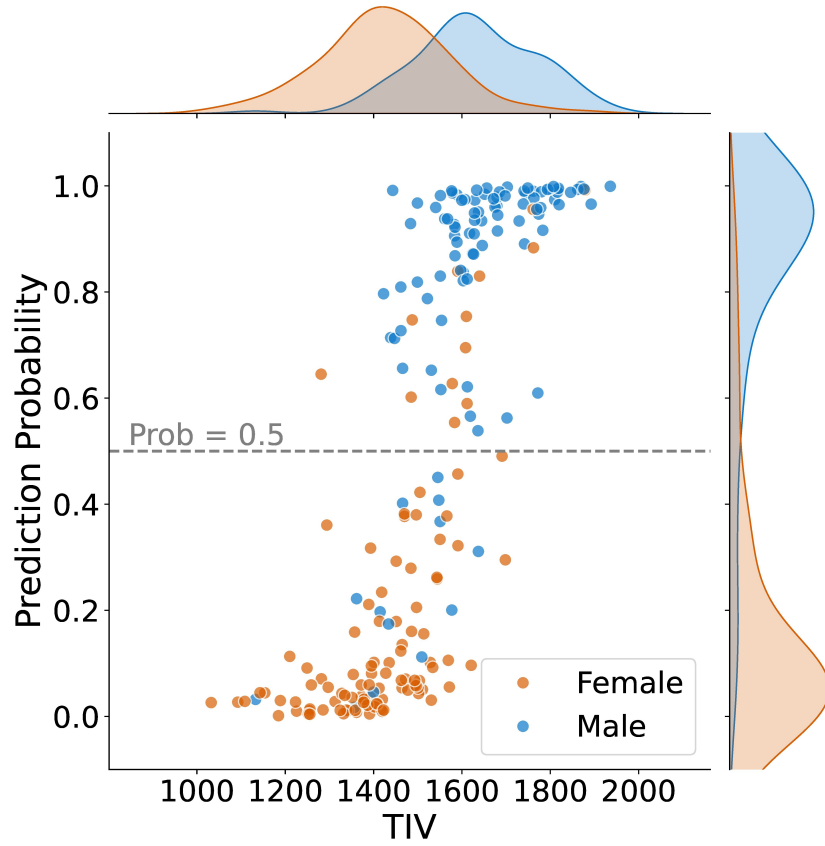


Figure 4.4: Sex classification prediction according to total intracranial volume (TIV) for 211 samples, including density plots for male and female distributions. The dashed line represents the classification threshold at a probability of 0.5. While the extremes of the TIV scale are strongly correlated with sex prediction—higher for males and lower for females—there is less bias in the middle TIV range.

Using KDE with a 20% threshold, we identified the overlapping TIV range of 1378 to 1666 (R_{tiv}). The model’s classification performance was most balanced between sexes in this range. The model’s accuracy within this range suggests that the reliance on TIV as a discriminative feature was minimized, resulting in more equitable performance across sexes.

In order to create the final averaged saliency map, a balanced subset was extracted from R_{tiv} , comprising an equal number of male and female samples (98 in total). This subset was specifically chosen to include only correctly classified samples. This approach ensures the final map is equally influenced by female and male samples and represents reliable and consistent regions considered important. We computed a final averaged saliency map from this subset, depicted in Figure 4.5.

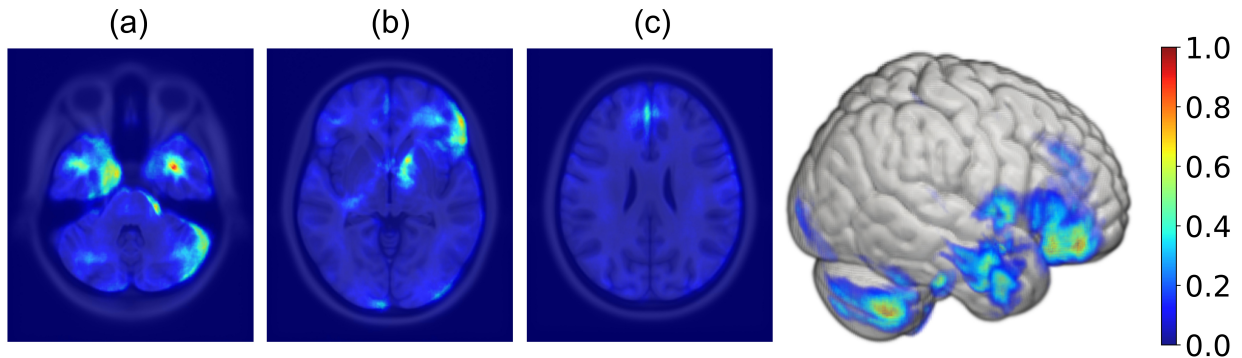


Figure 4.5: Axial MRI slices displaying average saliency map for 98 correctly classified samples (50% females, 50% males) in the TIV range of 1378 to 1666, with blue to red indicating low to high importance. The range was determined using KDE, filtering out values below 20% of maximum density for both sexes. From left to right, Panels (A), (B), and (C) highlight different axial levels with key brain structures for sex classification.

Supratentorial regions that were most important for prediction included the inferior frontal lobe and frontal operculum (seen in panel B), the parasagittal frontal lobe, more specifically the medial aspect of the superior frontal gyrus (seen in panel C), the mesial temporal lobe cortex and subcortical white matter (seen in panel A), the anterior limb and genu of the internal capsule (seen in panel A) and the optic chiasm and adjacent posterior aspect of the gyrus rectus (not shown). Infratentorial regions that were most important for prediction included the anterior and lateral surface of the pons and mesencephalon (seen in panel A) and the cerebellar hemispheric surfaces, particularly the lateral surface (seen in panel A).

To quantify the importance of brain regions, we computed saliency scores, which represent the percentage of salient voxels within each labeled region from the CerebrA atlas. Additionally, we calculated weighted saliency scores, which incorporate both the proportion of salient voxels and their saliency values, reflecting the significance of each region in the model’s predictions. Detailed results for all regions are provided in the appendix. The top 10 regions identified through this analysis, ranked by weighted saliency scores, include the left entorhinal cortex, right pars orbitalis, right pars

triangularis, right pallidum, right basal forebrain, right optic chiasm, left rostral anterior cingulate, right pars opercularis, right entorhinal cortex, and right lateral orbitofrontal. Figure 4.6 visualizes the saliency map overlaid on these top anatomical regions identified by the model along with their corresponding weighted saliency scores.

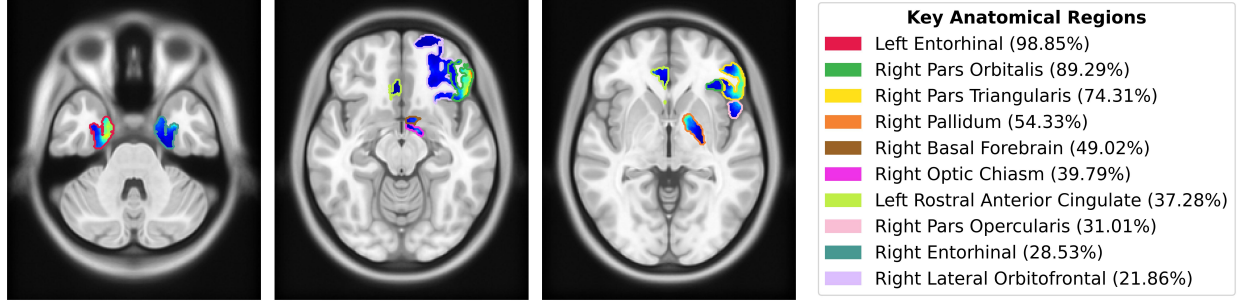


Figure 4.6: Saliency map overlaid on the brain atlas, highlighting the top 10 anatomical regions from the CerebrA atlas identified by the model using weighted saliency scores. Weighted saliency scores represent each region’s importance in the model’s predictions, considering both the proportion of salient voxels and their saliency values. The regions are color-coded according to the legend, with percentages indicating the weighted saliency score for each region.

4.4 Discussion

The proposed sex classification model achieved an accuracy of 87% on the test set. We deliberately chose not to apply TIV adjustment techniques, such as TIV matching, which often results in the exclusion of a significant portion of the available dataset. Additionally, we did not apply preprocessing steps like non-linear spatial and Z-Score normalization. By minimizing the use of preprocessing, we aimed to keep the brain structures and features as close to their natural form as possible, avoiding the introduction of artifacts or noise into the images.

While TIV correction is a common practice, it may not always be necessary or desirable depending on the objectives of the study. Given our goal of developing models that are more generalizable and less dependent on specific preprocessing steps, avoiding TIV correction was a considered and valid choice. In the post-hoc analysis of TIV, a comparison of model performance across TIV values indicated that predictions were less biased in R_{tiv} where the TIV distribution of females and males had the most overlap (Figure 4.4). The model’s accuracy in R_{tiv} was comparable to its overall accuracy. Moreover, the averaged saliency maps over all test samples (Figure 4.3) and the balanced

subset in R_{tiv} (Figure 4.5) showed similar regions with a Spearman correlation of 0.99 ($p < 0.05$). Figure 4.7 further confirms consistent predictive regions across TIV quantiles and sexes.

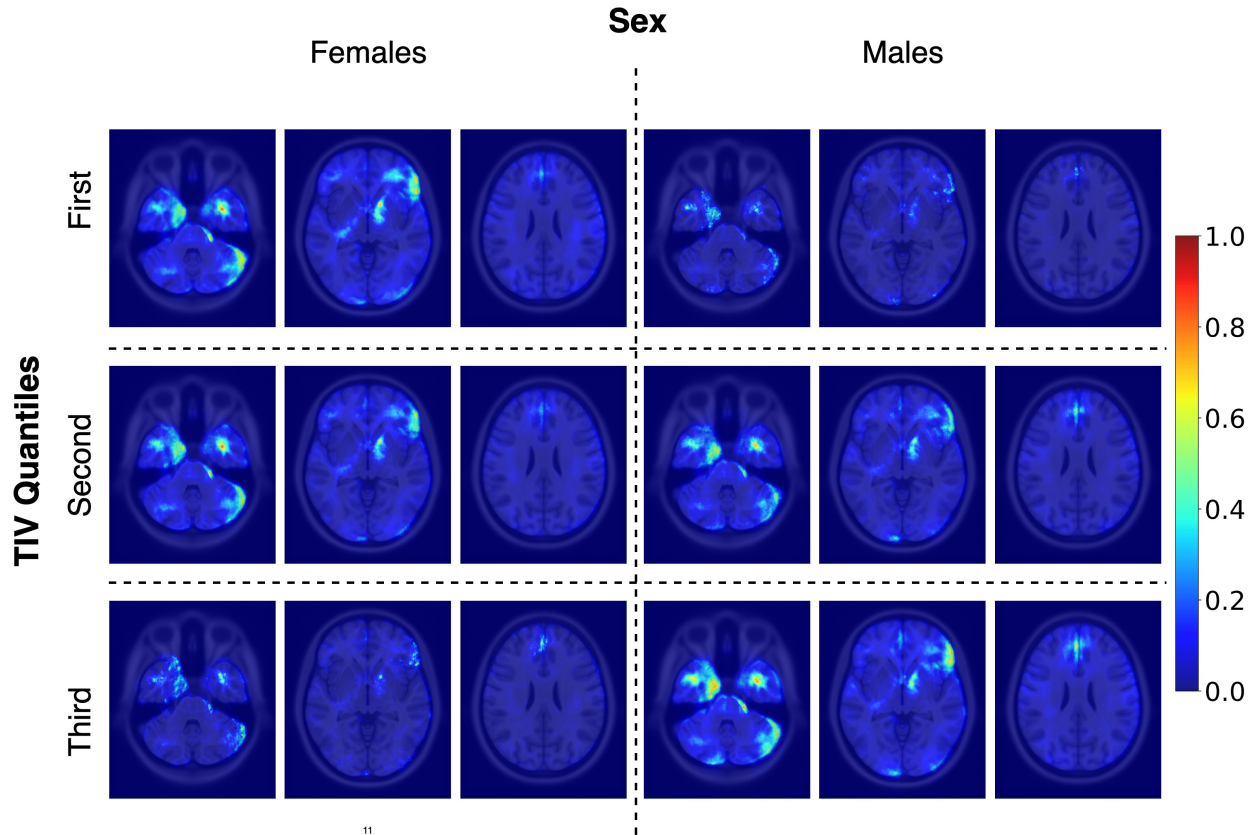


Figure 4.7: Average saliency maps for correctly classified females and males across three TIV quantiles, each representing a specific range of TIV values. Panels (A), (B), and (C) showcase axial levels that highlight key brain structures relevant for sex classification.

This study advances sex classification using brain MRI data by employing a 3D CNN model with minimal preprocessing, intentionally avoiding TIV adjustments to maintain natural brain structure variations. Unlike studies such as [Ebel et al. \(2023\)](#) and [Wiersch et al. \(2023\)](#), which control for brain size to reduce bias, our approach embraces these variations to explore their relevance in clinical settings. [Ebel et al. \(2023\)](#) used logistic regression on volume-matched gray matter data, achieving 92% accuracy, and a 3D CNN (BraiNN) on the same data, achieving 88% accuracy. [Wiersch et al. \(2023\)](#) utilized voxel-wise gray matter volume with an SVM, reaching 92.55% accuracy after TIV control. In contrast, [Hu et al. \(2019\)](#) focused on optimizing their multi-layer 3D convolutional extreme learning machine (MCN-ELM), achieving 98.06% accuracy without TIV adjustment, but they did not emphasize explainability or the identification of brain regions contributing to sex

differences.

Our minimal preprocessing allows the identification of nuanced sex differences, validated through interdisciplinary radiological assessment. While [Bozek et al. \(2021\)](#) also utilized explainability using SHAP analysis on volumetric features, our approach extends this by leveraging 3D MRI images to create a detailed “Sex Differences Map” through saliency analysis. This not only provides a high classification accuracy of 87% but also offers a robust, interpretable map of sex differences, validated by experts. To our knowledge, no other study has focused as extensively on visualizing important brain regions for sex classification.

Upon reviewing the saliency maps, Radiologist J. Ospel (co-author) observed that key regions for sex prediction in our model, such as the mesial temporal lobe cortex and superior frontal gyrus cortex, are situated in the grey matter. This observation aligns with previous studies showing notable differences in grey matter volumes between males and females, even when total brain volume is accounted for ([Liu et al., 2020](#)). Grey matter predominantly contains the somatodendritic tissue of neurons (i.e., the actual cell bodies), whereas white matter is primarily comprised of myelinated connecting axons (the “connecting fibers”). Some studies have shown a relatively higher percentage of grey matter in women, and it has been hypothesized that this results in prioritization of computational power over transfer across distant regions and could compensate for the overall lower brain parenchymal volume in women. Interestingly, not only total grey matter volumes, but also the distribution of grey matter between the cerebral hemispheres, have been shown to differ between women and men, with women showing a more symmetric distribution between both hemispheres ([Gur et al., 1999](#)).

Of note, regions like the cerebellum and temporal lobe cortex, known for their sex differences in grey matter volume ([Lotze et al., 2019](#)), were also significant in our model. Moreover, our model identified areas like the internal capsule and optic chiasm/posterior gyrus rectus, which, to our knowledge, have not been previously highlighted as being different among sexes in the literature. These regions have distinct functions: the internal capsule is crucial for connecting the cerebral hemispheres with various subcortical structures, the optic chiasm is part of the optic pathway and contains the crossing fibres of both optic nerves – lesions that compress the optic chiasm such as

pituitary adenomas, often lead to homonymous hemianopsia, i.e., visual field loss in the same halves of the visual field of each eye (Wolberg et al., 2024). Other regions, such as the inferior frontal gyrus, have more complex roles in inhibitory and attention control, and language production, depending on hemisphere dominance (Hampshire et al., 2010; Ishkhanyan et al., 2020).

Quantitative analysis using the CerebrA atlas further supported our findings. For each labeled region, we calculated a saliency score, representing the proportion of salient voxels, and a weighted saliency score that accounted for voxel intensity. These scores quantified the importance of each brain region in the model’s predictions. The top regions identified through this analysis included the left and right entorhinal cortex (part of the medial temporal lobe), right pars orbitalis, right pars triangularis, right pars opercularis, and right lateral orbitofrontal cortex (regions in the inferior frontal lobe), as well as the right optic chiasm. These findings align with the radiologist’s observations and provide a more detailed understanding of the neuroanatomical areas contributing to our model’s decision-making. Additionally, the quantitative analysis highlighted other regions, such as the right pallidum, right basal forebrain, and left rostral anterior cingulate (Figure 4.6).

Given the association of structural brain anatomy with functionality and behavioral patterns, one might hypothesize that sex differences in brain structure also translate into differences in functionality and behavior. However, this is not necessarily true. For example, while females have brain volumes that are on average 10-15% smaller compared to males, there is no difference in average intelligence levels between the two sexes (Burgaleta et al., 2012a). This highlights the fact that brain size or structure alone does not directly correlate with cognitive performance or behavior.

There is also growing recognition of the numerous compensatory mechanisms and redundancies in the brain that mitigate the impact of anatomical differences on function. Evolutionary processes, hormonal influences, and developmental factors all contribute to shaping brain structure, but these factors do not always lead to functional disparities (DeCasien et al., 2022a). Therefore, caution should be taken when inferring sex differences in functionality and behavior based solely on structural differences, as observed anatomical variations may not necessarily translate into functional or behavioral differences between men and women.

We assessed the model’s generalization by aggregating multiple public datasets from various

institutions. This strategy increased data volume and introduced a range of patient demographics and clinical practices, enriching our dataset’s diversity. Including different vendors and magnetic field strengths further diversified our dataset, which is crucial for developing robust DL models adaptable to various clinical settings. Table 4.2 confirms this, showing consistent performance across subgroups for each variable and reinforcing the model’s reliability across diverse clinical scenarios. These measures collectively boost accuracy and reliability, improving real-world healthcare applicability (Rouzrokh et al., 2022).

Sex differences in brain structure are known to interact with aging, which can influence the performance of classification models across different age groups. Research has shown that while sex-related structural differences are present throughout adulthood, they may be more prominent in early to mid-adulthood (Lotze et al., 2019). We analyzed model performance across different age ranges using thresholds based on key periods of brain aging identified in the literature. The first threshold was set at 55 years, as studies indicate that this is the period when more pronounced age-related changes in brain structure, particularly in gray and white matter, begin to emerge, marking the transition from midlife to older adulthood (Raz et al., 2005). The second threshold was set at 70 years, reflecting the transition into advanced aging, where the rate of brain atrophy accelerates significantly, particularly in regions such as the prefrontal cortex and hippocampus (Zhang et al., 2010). These thresholds allow us to explore how aging might impact sex classification model performance, capturing both midlife transitions and advanced aging effects. As shown in Table 4.2, the model performed consistently across all age groups, with balanced accuracy values of 0.889 for individuals under 55, 0.844 for those between 55 and 70, and 0.886 for those over 70. While there is some variation in performance, the model’s ability to classify sex is generally stable across these age ranges. The slight dip in performance in the 55-70 age group could reflect the onset of age-related changes that modestly affect the structural features the model relies on, but the overall consistency suggests that sex classification remains effective across different age ranges.

Many brain disorders have a clear sex prevalence. As an example, demyelinating diseases such as multiple sclerosis and other immune-mediated inflammatory brain disorders are more prevalent in females (Avila et al., 2018), while certain types of neurodegenerative diseases, such as Parkinson’s disease, occur more commonly in males (Young et al., 2023). This implies that the pretest probability

for many neurological conditions (the probability of a screened person having the disease ([Attia et al., 2004](#))) is influenced by patient sex, and accurate classification of patient sex would therefore be expected to improve any algorithm aimed at diagnosing this condition. Furthermore, for some brain conditions, the prognosis differs between males and females. For example, after being diagnosed with glioblastoma, the most aggressive of the primary brain tumors, males have, on average, shorter survival compared to females ([Moore et al., 2022](#)).

Furthermore, sex differences in brain atrophy patterns offer another example of how these variations might inform individualized treatment strategies. For instance, for Alzheimer’s disease, women often experience a faster atrophy rate in key brain regions compared to men, even with similar biomarker levels. These morphological differences could help tailor treatments more precisely for men and women, moving toward a more personalized medicine approach in AD ([Ferretti et al., 2018](#)). Incorporating sex-related information into diagnostic and prognostic models would improve the tool’s reliability, supporting fairer and more effective treatment strategies.

To enhance bias mitigation and model optimization using the sex-differences saliency map, we propose two strategies. First, for post-processing bias mitigation, we suggest evaluating model performance through an explainability map on the test set. If this map shows a strong correlation with the sex-differences saliency map, it could indicate the presence of sex bias within the model. Recognizing such biases is crucial for initiating corrective actions. Second, for in-processing bias mitigation during training, we recommend adapting the reweighing bias mitigation algorithm ([Kamiran and Calders, 2012](#)) at a voxel level. This involves adjusting voxel weights in images based on the sex-differences saliency map to direct the model’s focus toward more task-relevant brain features and reduce the impact of sex-specific characteristics, thereby promoting a more equitable predictive process.

Our study has limitations, particularly regarding the distinction between sex and gender in medical research. Sex, categorized as male or female, refers to biological factors and is considered static. In contrast, gender encompasses social roles, behaviors, expressions, and identity, existing along a continuum and subject to change over time. While gender identity usually aligns with biological sex, it’s important to note that transgender individuals, though a small percentage ($< 1\%$

in the U.S. (Chan, 2019)), are part of the population. Therefore, some differences identified by our algorithm might be attributable to gender rather than sex. Additionally, the included datasets presented self-reported sex information, which, for a few individuals, may reflect gender identity rather than sex assigned at birth. This practice is common in biomedical research, where self-report questions are often used to characterize sex or gender identity, and the distinction between these concepts is not always made clear (Stites et al., 2023). This lack of clarity can introduce variability in datasets, as research participants may reference different concepts when answering questions about sex or gender. While we do not expect this issue to significantly impact the development of our algorithm, it is a potential source of variability that should be acknowledged.

Another limitation of our study is related to the class saliency map method. While this method provided the best results in our experiments and the generated maps were validated by our radiologist, it is important to acknowledge that it is sensitive to noise, which can result in less stable visualizations. The effectiveness of this method is closely tied to the CNN architecture, potentially limiting its ability to fully capture non-linearities in deeper models.

Lastly, although we employed stratified splitting in our study to ensure representativeness, we recognize that the use of stratified cross-validation could further improve the assessment of our model’s generalizability. Furthermore, we acknowledge that not employing Intensity Standardization techniques (Nyúl et al., 2000) may limit our ability to fully correct for inter-scanner variability in intensity distributions. We applied z-score normalization; however, this approach does not account for the non-standardness inherent in MRI intensities across different scanners, which can result in residual discrepancies affecting comparability and generalizability. Incorporating intensity standardization in future work could help address these limitations and improve consistency across datasets.

4.5 Chapter Conclusion

A key outcome of our study is the development of sex-differences maps through saliency analysis, which identified critical brain regions involved in sex classification. These maps offer DL-based insights into sex differences in the brain and hold the potential to enhance diagnostic and prognostic

algorithms by identifying and addressing sex-based biases, ultimately contributing to more equitable healthcare interventions. This interdisciplinary approach, combining ML and radiological expertise, emphasizes the importance of fairness in ensuring that the outcomes are not only accurate but also reliable and relevant. Our sex classification model achieved an accuracy of 87%, aggregating four diverse datasets with minimal preprocessing to maintain brain structure integrity. The model showed biases at extreme brain sizes but was less biased in ranges where male and female Total Intracranial Volume distributions overlapped the most.

The insights gained from the sex-differences maps also have broader applicability. For example, these maps could be utilized for the refinement of diagnostic tools for neurological conditions for both males and females by informing critical brain regions. Additionally, they could be adapted to other fields where understanding region-specific influences is important, such as targeted medical interventions based on brain imaging data. While promising, our approach has certain limitations, including the sensitivity of saliency maps to noise, which may affect the stability of the visualizations. Future work could focus on refining these maps to enhance their robustness and exploring their application in different contexts to further validate their utility.

Chapter 5

Conclusions

5.1 Key Findings

This thesis aims to investigate the influence of sex-related differences in brain MRI on deep learning models, with a particular focus on both their impact on predictive performance and their underlying neuroanatomical basis. The first study contributes to these aims by examining how sex-specific characteristics in brain MRI, combined with different data-handling strategies, influence a downstream task such as brain age prediction. The second study advances this investigation by exploring which brain regions underpin these sex differences from a deep learning perspective. This was achieved by training a sex classification model and employing explainability techniques. These complementary investigations enabled us to isolate morphological cues tied to sex-based variation rather than merely noting how such variation influences downstream performance. Minimal data preprocessing was a central design principle to preserve subtle anatomical signals and minimize confounding effects of extensive data processing; Therefore, we only used brain extraction, rigid registration, and intensity normalization. This approach clarified how demographic variations, particularly sex, can emerge as critical features in automated analyses of brain MRI.

Our exploration of brain age prediction (Chapter 3) revealed that subtle changes in dataset composition, such as training models on female-only or male-only subsets, could shift which cortical or subcortical regions the model relied on. These findings align closely with existing literature on sex differences in brain aging trajectories [Sanford et al. \(2022\)](#), underscoring that traditional

brain-age pipelines may obscure or dilute important sex-specific patterns by treating diverse datasets as homogeneous. Furthermore, despite similar overall performance metrics, explainability maps revealed that models trained on different development subsets learned distinct neuroanatomical features. This reinforces the critical role of explainability techniques in thoroughly validating model outcomes and ensuring robustness against latent biases.

Our sex classification work (Chapter 4) demonstrated that even a straightforward convolutional neural network, trained under minimal constraints, could reliably detect sex-based morphological differences in various brain regions, including temporal and frontal lobes and the cerebellum. The accuracy of this model was comparable to prior studies (Ebel et al., 2023; ?), and the identified neuroanatomical regions closely correspond to those previously reported in the literature as influenced by sex (Lotze et al., 2019). The subsequent explainability analyses led to the construction of a “Sex Differences Map”, revealing localized markers of sex that transcended mere volumetric distinctions. Notably, our results confirm observations from Ebel et al. (2023) that extremes in TIV can overshadow subtler, clinically relevant signals. By specifically examining mid-range TIV values, where male and female distributions overlap the most, we demonstrated that the model relied less on overall brain size and more on finer precise anatomical distinctions. This range allowed for a more balanced and equitable evaluation, reducing the confounding influence of TIV extremes that otherwise skew sex classification.

Overall, these findings highlight the importance of avoiding overly aggressive preprocessing and confirm that clinically meaningful demographic distinctions can be discerned more accurately in datasets where natural variability is respected.

5.2 Impacts

By bringing sex-based analyses to the forefront and devising a minimal-preprocessing design that highlights morphological nuances, this thesis makes methodological and conceptual contributions to the broader field of deep learning for neuroimaging. In showcasing how sex influences model behavior—both in age estimation and binary classification—we underscore the importance of recognizing demographic subgroups in automated pipelines, a key consideration for bias-sensitive domains like

healthcare. Our integrative approach, which combines direct assessment of morphological signals with explainability techniques, opens pathways for more transparent, equitable, and scientifically informed uses of deep learning. As research in AI-driven neuroimaging continues to evolve, our findings reinforce the need to further investigate and consider intrinsic biological variation rather than normalize it away, moving toward promoting more accurate and fair clinical implementations. Additionally, collaboration with a radiologist proved vital in validating and contextualizing our explainability findings, offering a clinical perspective on the highlighted regions and guiding how AI-driven insights might translate into enhanced patient care.

Additionally, this thesis contributes to the principles of open science by leveraging publicly available neuroimaging datasets and ensuring that our methodologies and code are accessible through an open-source repository on GitHub. By making our codebase publicly available, we promote transparency, reproducibility, and collaboration within the scientific community. This aligns with the broader movement toward open and fair AI in healthcare, where accessibility and shared resources enable researchers to build upon existing work, validate findings, and further advance the field.

5.3 Limitations and Future Directions

Although we integrated multiple large-scale datasets, we nonetheless encountered challenges in achieving truly balanced demographic representation. Certain subpopulations were underrepresented, and site-level inconsistencies, such as different scanners or field strengths, constrained the uniformity of our analyses. Moreover, restricting ourselves to rigid registration preserved gross volumetric signals but may have limited the precise alignment of cortical features that advanced registration methods can capture. Despite these issues, our methodological approach still demonstrated that explainability-focused strategies, combined with minimal preprocessing, can uncover distinct sex-related features in otherwise diverse neuroimaging datasets, thus offering a practical foundation for further methodological refinements.

Looking ahead, we envision utilizing the “Sex Differences Map” more directly as a mechanism for bias identification and mitigation, particularly in disease classification tasks where demographic confounders may obscure pathological signals. For bias identification, particularly in cases where

sex metadata is unavailable, the sex differences map could serve as a reference for analyzing feature reliance in new downstream tasks. By generating an averaged saliency map for a model trained on a different task—such as Alzheimer’s disease classification—and computing its correlation with the sex differences map (either globally or within specific sex-associated regions), we could assess the extent to which the model depends on sex-related features. A high correlation could indicate unintended reliance on sex-specific brain structures, serving as a starting point for further investigation and bias mitigation strategies.

For bias mitigation, two potential approaches could be explored: (1) masking the top sex-related regions in the input images to prevent the model from using them, or (2) applying a weighting mechanism that assigns lower importance to sex-related regions, thereby reducing their impact on the loss function during training. Both methods could help ensure that models prioritize disease-relevant features rather than demographic differences.

It is important to note that the applicability of the “Sex Differences Map” may also be influenced by architectural and methodological choices. Specifically, our CNN architecture inherently affects the generalizability of the Sex Differences Map. While our findings were robust within the selected CNN architecture, the consistency and generalizability of identified features might differ with alternative or deeper network architectures. Furthermore, the selected explainability method (saliency maps) is sensitive to noise, potentially impacting the stability and reliability of visualizations across different models or explainability approaches. To address these concerns, future research should systematically explore how variations in model architecture and explainability methods influence the reproducibility and stability of the Sex Differences Map.

To further refine the “Sex Differences Map” and enhance its generalizability, future work could focus on using larger and more demographically diverse datasets to improve robustness across populations. Additionally, experimenting with various explainability techniques—such as integrated gradients, or layer-wise relevance propagation—or combining multiple methods could provide a more comprehensive and reliable representation of sex-related features. However, one of the key challenges in this process is the inherent differences between explainability methods, which can make direct comparisons difficult. Developing statistical frameworks to systematically compare maps

generated by different techniques would help standardize interpretations and increase confidence in the insights drawn from these maps. Addressing these challenges will be crucial for establishing the sex differences map as a reliable tool for bias detection and mitigation in AI-driven neuroimaging applications. Lastly, beyond sex-related analyses, future work could extend this approach to other relevant demographic variables to further enhance fairness in AI-driven neuroimaging.

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Appendix

Unweighted and weighted saliency scores are presented for each brain region in the averaged sex differences saliency map for 98 correctly classified samples (50% female, 50% male) within a TIV range of 1378 to 1666 ml. These brain regions were identified from the CerebrA atlas.

Region Name	Saliency Score (%)	Weighted Saliency Score (%)
Left Entorhinal	98.85	98.85
Right Pars Orbitalis	91.76	89.29
Right Pars Triangularis	85.19	74.31
Right Pallidum	72.02	54.33
Right Basal Forebrain	87.50	49.02
Right Optic Chiasm	71.26	39.79
Left Rostral Anterior Cingulate	67.40	37.28
Right Pars Opercularis	57.70	31.01
Right Entorhinal	59.94	28.53
Right Lateral Orbitofrontal	52.87	21.86
Right Inferior temporal	46.92	20.09
Left Optic Chiasm	42.48	18.99
Right Accumbens Area	46.65	18.28
Right Lateral Occipital	41.65	14.71
Left Inferior temporal	32.53	12.38
Right Cerebellum Gray Matter	32.63	11.52
Left Amygdala	32.97	11.19

(continued)

Region Name	Saliency Score (%)	Weighted Saliency Score (%)
Right Insula	36.80	10.84
Left Superior Frontal	32.66	10.47
Left Parahippocampal	33.12	10.28
Right Putamen	29.72	10.26
Left Inferior Lateral Ventricle	31.35	9.75
Left Caudal Anterior Cingulate	30.28	9.37
Left Cerebellum White Matter	29.15	8.99
Left Pars Triangularis	27.57	8.27
Left Cerebellum Gray Matter	27.73	7.95
Right Superior Temporal	27.25	7.82
Right Rostral Anterior Cingulate	26.72	7.61
Left Medial Orbitofrontal	24.19	7.55
Left Brainstem	26.74	7.33
Right Brainstem	24.02	7.05
Left Ventral Diencephalon	23.70	6.27
Left Hippocampus	20.81	5.83
Left Basal Forebrain	18.23	5.55
Right Rostral Middle Frontal	20.04	5.05
Left Superior Temporal	18.59	4.99
Right Caudate	20.82	4.92
Left Pericalcarine	20.50	4.86
Left Lateral Occipital	18.73	4.80
Left Paracentral	17.66	4.75
Right Cerebellum White Matter	16.22	4.11
Left Lateral Orbitofrontal	16.33	4.10
Right Middle Temporal	16.25	4.04

(continued)

Region Name	Saliency Score (%)	Weighted Saliency Score (%)
Right Thalamus	16.76	3.76
Right Medial Orbitofrontal	14.77	3.68
Right Superior Frontal	15.39	3.58
Left Fusiform	15.54	3.44
Left Pars Orbitalis	12.48	3.05
Left Middle Temporal	12.69	2.85
Left Third Ventricle	12.07	2.57
Right Amygdala	11.27	2.48
Right Ventral Diencephalon	11.66	2.40
Left Pars Opercularis	10.02	2.19
Left Transverse Temporal	10.84	2.13
Left Precuneus	8.89	1.61
Right Precentral	8.42	1.44
Right Fusiform	6.69	1.29
Right Caudal Anterior Cingulate	7.04	1.23
Left Lingual	6.96	1.21
Left Lateral Ventricle	8.76	1.17
Left Accumbens Area	5.97	1.14
Right Postcentral	5.34	0.88
Left Rostral Middle Frontal	4.42	0.87
Left Caudate	5.58	0.81
Left Thalamus	4.98	0.79
Right Lateral Ventricle	5.59	0.73
Left Cuneus	3.52	0.58
Left Pallidum	2.12	0.43
Left Isthmus Cingulate	3.19	0.42

(continued)

Region Name	Saliency Score (%)	Weighted Saliency Score (%)
Right Pericalcarine	2.13	0.38
Left Postcentral	2.41	0.36
Right Lingual	1.97	0.31
Right Inferior Parietal	1.94	0.31
Right Superior Parietal	1.67	0.28
Right Inferior Lateral Ventricle	1.98	0.23
Right Hippocampus	1.73	0.20
Right Third Ventricle	1.18	0.16
Left Putamen	1.16	0.13
Left Precentral	0.68	0.11
Left Insula	0.87	0.10
Right Supramarginal	0.57	0.08
Right Precuneus	0.42	0.06
Left Fourth Ventricle	0.33	0.04
Right Paracentral	0.23	0.04
Right Vermal lobules VIII-X	0.19	0.03
Left Inferior Parietal	0.20	0.02
Right Fourth Ventricle	0.13	0.01
Left Superior Parietal	0.07	0.01
Left Posterior Cingulate	0.02	0.00
Left Caudal Middle Frontal	0.01	0.00
Right Caudal Middle Frontal	0.00	0.00
Left Vermal lobules VI-VII	0.00	0.00
Left Vermal lobules I-V	0.00	0.00
Right Vermal lobules VI-VII	0.00	0.00
Right Transverse Temporal	0.00	0.00

(continued)

Region Name	Saliency Score (%)	Weighted Saliency Score (%)
Right Vermal lobules I-V	0.00	0.00
Left Vermal lobules VIII-X	0.00	0.00
Right Cuneus	0.00	0.00
Right Parahippocampal	0.00	0.00
Right Posterior Cingulate	0.00	0.00
Right Isthmus Cingulate	0.00	0.00
Left Supramarginal	0.00	0.00